UNIVERSITY OF COIMBRA

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Department of Informatics Engineering



Project Management

Project Documentation

OliveTrack

Requirements and Architecture

Olive Harvest Management Platform

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1 Introduction

1.1 Purpose of the Document

This document defines the proposed requirements and architecture of OliveTrack, a platform for managing olive harvest operations.

It establishes a shared understanding of the problem, intended users, functional scope, business rules and expected system behaviour. It also connects the use cases to the functional requirements, quality requirements and architectural decisions developed by the team.

The document supports both product development and project management. An explicit description of the expected behaviour helps the team identify missing work, define implementation and validation tasks, allocate responsibilities and prepare effort estimates.

This document complements the Statement of Work and the other artefacts required by the Project Management course. It does not independently amend the agreed project boundaries or replace the project's work plan, task assignments or acceptance records.

1.2 Project Context

Olive harvest management involves information that is produced by different people and becomes available at different times.

Before fieldwork, the owner needs to decide which workers are expected, where they will work and, where known, their schedules and tasks. During or after the working day, actual attendance, work periods, tasks and breaks must be recorded.

Harvested olives are transported and delivered to a cooperative or another receiving destination. Weighing tickets provide information about the delivered load. Analysis results and monetary values may arrive later.

Worker documents, including contracts and insurance records, must also be accessible. Some documents require a signing interaction. Insurance information is particularly important when work takes place in the field.

When this information is distributed across paper records and separate files, it becomes harder to retrieve, reconcile and summarise the campaign's results. Differences between planned and actual work, missing analysis results and ambiguous weight values can also lead to incorrect conclusions.

OliveTrack proposes a shared digital environment for the owner's business. It supports properties with optional parcels, harvest campaigns covering one or more areas, worker records, operational entry, document consultation and campaign indicators.

Field connectivity may be unreliable. Essential field operations therefore require prepared offline access and local entry, followed by synchronisation when connectivity returns. Administrative operations may require connectivity, as specified in the individual use cases.

The project is being developed within the Project Management course. The proposed scope must remain compatible with the academic schedule, available resources and agreed delivery objectives.

1.3 Objectives

The proposed solution pursues the following objectives:

1. **Organise the harvest context.** Maintain identifiable properties, optional parcels and

campaigns without mixing operational records from different areas or harvest periods.

2. **Separate planning from actual work.** Support daily and multi-day planning, while requiring explicit confirmation of attendance and actual work.
3. **Record work consistently.** Distinguish machinery operation from general work, account for explicitly recorded breaks and prevent overlapping actual periods for the same worker.
4. **Track deliveries and source information.** Connect transport, delivery, weighing, analysis and monetary information without inventing missing measurements or applying deductions twice.
5. **Support worker-document workflows.** Provide controlled access to uploaded documents, prepared offline insurance consultation and the agreed signing interaction.
6. **Provide understandable indicators.** Present campaign results with clear units, reporting scope, data completeness and provisional or final status where applicable.
7. **Support work in the field.** Allow essential operations to continue with prepared authorised data when a network connection is unavailable.
8. **Support project planning and validation.** Provide a traceable basis for requirements, implementation tasks, estimates, acceptance checks and team review.

Detailed measurable targets are addressed in the non-functional requirements. The objectives above do not constitute verified performance or operational results.

1.4 Sources and Document Status

The current document draws on:

- The *OliveTrack Team Guide*, dated 9 October 2026, describing the business context, illustrative examples and unresolved questions.
- The existing Statement of Work, persona descriptions, general architecture, data architecture and non-functional requirements proposals supplied to the team.
- Initial responses supplied by the teammate acting as an intermediary with the client.
- The subsequent requirements review, including the use-case discussions and proposed resolutions recorded during document development.

These sources do not all have the same approval status. An illustrative example, team preference or design assumption must not be presented as a client-approved business rule.

The document is currently a working draft. The use cases consolidate the team's current understanding and proposals. They remain subject to team review and client validation where necessary.

The pending decision register in Chapter 10 identifies the unresolved scope, business and implementation questions and the proposed roles needed for validation.

Conflicting source statements are resolved through an explicit decision, rather than silently selecting one interpretation.

Use cases and rules cite the decision IDs; complete question wording, status and proposed validation responsibilities are maintained in Chapter 10.

2 Users, Personas and Permissions

2.1 Stakeholders

OliveTrack affects people who use the application, people whose information is managed and parties that supply information or assess the project.

Farm Owner and Client. Explains the operational needs, validates business behaviour and uses the application to manage the harvest and inspect its results.

Field Manager. A potential delegated operational user. Its inclusion remains to be confirmed. If included, the manager acts within the permissions granted by the owner.

Workers. Their profiles, attendance, work periods and documents are managed. Workers may participate in signing without holding application accounts.

Cooperative or Receiving Organisation. Supplies weighing information, analysis results and monetary values. Direct access to OliveTrack is not assumed in the current scope.

Project Team. Analyses requirements, proposes the solution, implements and validates the agreed scope, and maintains the project documentation.

Course Faculty and Evaluators. Establish academic expectations and assess the project deliverables, management process and demonstrated results.

Business validation and academic acceptance are related but distinct. Client agreement does not replace the course's assessment requirements.

2.2 User Personas

The profiles below describe working assumptions for requirements discussion. They represent roles and usage contexts; they are not evidence of a completed user study.

2.2.1 Farm Owner

Role and context. Responsible for the business's properties, workers and harvest decisions. May prepare administrative information with connectivity and consult or record operational information in the field.

Main goals.

- Organise properties, parcels and campaigns.
- Prepare work without repeatedly entering the same information.
- Understand who attended and what work was actually performed.
- Access the correct worker documents.
- Follow deliveries and later results.
- Review campaign totals and investigate their supporting records.

Main difficulties. Information arrives at different times, manual records can be fragmented, and field connectivity is unreliable.

Design implications. Use clear business terminology, group entry with individual adjustments, explicit save states and understandable summaries with access to detail.

2.2.2 Field Manager

Status. Conditional persona, pending confirmation of the need for a separate application role.

Role and context. May coordinate workers and field operations on behalf of the owner. Uses the application within an explicitly authorised operational scope.

Main goals.

- Consult prepared schedules.
- Record attendance and actual work.
- Record transport and delivery information.
- Access relevant worker insurance during field operations.

Access limitations. Work-entry permission does not automatically include correction rights, contract access, monetary information or configuration rights.

Design implications. Prioritise essential field tasks, prepared offline information and clear permission boundaries.

2.2.3 Worker as a Participant

Role and context. A person whose work and documents are managed. May review and sign a document in person on the owner's or authorised manager's device.

Main needs.

- Be correctly identified in records.
- Review the document intended for signing.
- Review, redraw or cancel before confirming a signature.
- Avoid unintended access to other workers' information.

No worker self-service application or management account is assumed.

2.3 System Actors

The formal actor name is *Farm Owner*; running text uses *owner*.

Farm Owner. Primary application actor. Performs authorised management, operational and consultation tasks within the owner's business.

Field Manager. Conditional application actor with delegated permissions.

Worker. Participating actor in the signing interaction. Does not manage attendance or business information through an account.

The cooperative is currently an information source, rather than a direct application actor. Likewise, identifying a worker as a driver does not create another application role.

The owner and manager are user roles. A worker record is a business record. These concepts must not be treated as interchangeable identities.

Visitor. A Visitor is a person who is not authenticated. Visitors may register as owners, verify their email, log in or initiate account recovery. They cannot access business information.

Authenticated account holders. An owner holds an application account associated with the business created through registration.

If the Field Manager role is approved, a manager receives business access through an owner invitation and acts within the assigned scope and permissions.

Worker records are operational business records. Creating a worker record does not create an account. Workers review and sign documents through an authorised account holder's device without logging in themselves.

2.4 Access Permissions

Permissions are evaluated within the authorised business context and, where applicable, the relevant properties, campaigns, workers and documents.

The same person may have independent worker records in different businesses. Those records are not automatically merged or exposed to another owner.

The following matrix summarises the proposed access principles.

Consultation, creation, modification and correction permissions are distinct. A permitted interface action must also be authorised by the server.

Offline access is limited to previously prepared authorised information. Changes to permissions must be rechecked when the device reconnects. The detailed access-expiry and revocation policy remains to be defined.

The complete permission matrix must be reviewed if the Field Manager role is confirmed.

Account and access administration. Self-registration establishes ownership of a newly created business workspace. Email verification is required before active access is granted.

Only the owner may administer manager invitations, responsibility scopes and permissions under the current role model.

Authentication, business membership and operation-specific permissions are separate checks. A valid account does not grant access to another owner's business.

Manager permissions distinguish consultation, modification, correction, document operations and financial access. Insurance consultation is mandatory within an assigned worker-responsibility scope.

Financial information remains restricted to authorised online access.

The Field Manager role and its responsibility-scope granularity remain subject to stakeholder confirmation.

Capability	Farm Owner	Field Manager, if included
Manage properties, parcels and campaigns	Within the business scope.	Explicit permission required.
Manage worker profiles	Within the business scope.	Relevant consultation or editing permission required.
Plan and record work	Within the business scope.	Explicit operational permission required.
Correct confirmed work	Permitted under correction and campaign rules.	Separate correction permission required.
Consult worker insurance	Within the business scope.	Required for workers within the authorised operational scope.
Consult contracts and other documents	Within the business scope.	Specific document-type permission required.
Upload or update documents	Within the business scope.	Separate modification permission required.
Initiate worker signing	Within the business scope.	Explicit signing-initiation permission required.
Consult or edit monetary values	Within the business scope.	Specific monetary-data permission required.
Consult campaign indicators	Within the business scope.	Limited to authorised data and indicators.

3 Scope and Feature Overview

3.1 Included Scope

The proposed scope focuses on managing olive harvest operations within an owner's business.

It includes the following functional areas.

Properties and Parcels. Create and manage properties with optional internal parcels. Retain historical operational associations when descriptive information changes.

Harvest Campaigns. Organise campaigns covering whole properties or selected parcels. Manage planned and actual dates, closure, reopening and archival.

Worker Records. Maintain business-specific worker profiles, contact information and machinery capability. Support inactivation without automatically deleting historical records.

Work Planning. Prepare daily or multi-day plans, apply group values with individual adjustments, and reuse previous plans or actual work as proposals for future dates.

Attendance and Actual Work. Confirm daily attendance and actual assignments. Record task-specific periods, areas and optional breaks. Support authorised corrections.

Worker Documents. Upload, consult and update contracts, insurance and other relevant documents. Apply category-specific version rules and prepare insurance for offline access.

Worker Signing. Support the proposed in-person drawn-signature interaction, including prepared offline signing. Operational requirements remain subject to validation before deployment.

Transport and Deliveries. Record departure and delivery, registered drivers, vehicles, destinations and load provenance.

Weighing and Analysis. Manually record available ticket values, adjustments and analysis results. Preserve unknown information without inventing calculations.

Reported Monetary Values. Record the cooperative's current provisional or final amount for a ticket. Replace superseded values without counting them twice.

Campaign Indicators. Consult current work, production and permitted monetary indicators. Support property summaries, parcel detail and PDF export.

The use cases provide the detailed proposed behaviour. This overview groups capabilities without replacing their rules or acceptance criteria.

Account access. The proposed scope includes free owner registration, email verification, login, logout and password recovery.

Manager invitations and permission administration are conditional on confirmation of the Field Manager role.

Initial authentication and access administration require internet connectivity. Supported field operations may continue offline after prior authentication and preparation of the required data.

3.2 Excluded Scope

The current proposal excludes:

- Payroll execution, wage calculation, worker remuneration rates and payment processing.
- Invoicing, full accounting and commercial olive oil sales.
- Application-calculated price estimates for deliveries.
- Worker self-service accounts for attendance or business management.
- Direct cooperative or laboratory integrations.
- Automatic extraction of values from ticket images or documents.
- Independent transport-trip management beyond the delivery-associated transport record.
- Management of agricultural activities unrelated to the agreed olive-harvest scope.
- Automatic allocation of mixed-load weights or monetary amounts without approved rules.

The following are optional enhancements, rather than committed first-version capabilities:

- Map-based property and parcel boundaries, visual area selection and estimated map areas.
- Cross-campaign or cross-year comparisons.
- Detailed Excel exports.
- Dedicated document-printing workflows.

Document generation through UC-20 remains a candidate capability whose concrete scope must be confirmed.

3.3 MVP Proposal Pending Approval

The Minimum Viable Product has not yet been formally prioritised and approved.

The current proposed baseline is an end-to-end harvest workflow in which the owner can:

1. Prepare properties, campaign coverage, workers and necessary documents.
2. Plan work and consult prepared schedules.
3. Confirm attendance and record actual periods and tasks in the field, including without connectivity.
4. Record transport and delivery, then complete ticket, analysis and reported monetary information as it becomes available.
5. Consult relevant insurance, use the agreed signing interaction and inspect campaign results.

This baseline is a proposal for prioritisation, not a declaration that every drafted feature is equally essential or already feasible within the available resources.

Before estimates are finalised, the team and client must identify which capabilities are mandatory for academic demonstration and which are required for operational use.

In particular, operational signing, offline document protection and synchronisation recovery require explicit validation and cannot be considered complete solely because their interface is demonstrated.

Optional map features and unconfirmed document-generation requirements must not silently expand the baseline.

3.4 Feature Overview

Area	Main capability	Use cases
Account access	owner registration, email verification, login, logout and recovery.	UC-01–03
Manager access	Invitations, permissions and revocation, conditional on confirmation of the role.	UC-04
Land and campaign organisation	Properties, optional parcels and campaign lifecycle.	UC-05–06
Workforce	Worker profiles and machinery capability.	UC-07
Planning	Daily and multi-day plans, reuse and schedule consultation.	UC-08–09
Actual work	Attendance, task periods, breaks and corrections.	UC-10–11
Documents and signing	Document access, uploads, versions and worker signatures.	UC-12–14
Harvest logistics	Transport-associated deliveries, weighing tickets and analyses.	UC-15–17
Reported values	Current cooperative-reported ticket amounts.	UC-18
Campaign review	Indicators, supporting detail and PDF summaries.	UC-19
Document generation	Template-based generation, pending scope confirmation.	UC-20

Administrative configuration normally requires connectivity. Essential field entry and consultation use prepared authorised information and explicit synchronisation states.

Offline capability is defined per operation. It does not imply that the entire application or every historical record is available without a connection.

3.5 Assumptions and Dependencies

3.5.1 Business Assumptions

The current requirements review assumes that:

- Worker profiles and necessary documentation are prepared before fieldwork.
- A worker may perform different tasks and work in several areas during the same day.
- Campaigns may cover several properties or selected parcels.
- Source information may arrive after the physical delivery or campaign closure.
- One transport record is associated with each delivery in the first version.

- Each ticket and analysis belongs to one delivery. The reverse relationships remain subject to confirmation.
- Monetary values are supplied by the cooperative and associated with weighing tickets.

These assumptions must be revised when validated examples contradict them.

3.5.2 Business Dependencies

The team depends on client or intermediary input to confirm:

- Weighing-ticket terminology, deduction rules and source-value meanings.
- Analysis reference weights, result selection and yield calculations.
- Document types, templates and intended signing requirements.
- The Field Manager role and delegated permissions.
- First-version priorities and operational acceptance expectations.

Real examples should be provided with unnecessary personal information removed.

3.5.3 Technical Dependencies

The solution depends on:

- Devices and browsers capable of supporting the selected application approach.
- Successful preparation of authorised reference data before offline work.
- Sufficient local storage for pending operations and prepared documents.
- A central service for validation, persistence and synchronisation.
- Defined recovery behaviour for rejected operations, concurrent changes and interrupted transfers.
- Consistent deletion propagation for document versions removed under the retention rules.

The existing architecture proposes a PWA, local browser storage, a backend API and central relational persistence. The technical design must be reviewed against the consolidated requirements before being treated as final.

3.5.4 Project Dependencies

The academic schedule, team availability, hosting resources and required course deliverables constrain the implementation plan.

Use-case approval, task ownership, effort estimates and acceptance evidence must be tracked explicitly. The existence of a documented use case does not establish that its implementation has been completed or validated.

Identity and email delivery. Registration, invitations and password recovery depend on an email delivery mechanism and an identity-management implementation.

Business workspace. The current scope provides one business workspace per owner, containing the owner's properties, campaigns, worker records, documents and operational information.

Managing several independently owned business workspaces through the same owner account is outside the current baseline.

The Field Manager role and the permitted scope of delegated access remain subject to team and stakeholder confirmation.

Offline access depends on a previously authenticated session, protected local storage and a bounded offline authorisation period.

Session expiry, token lifetimes, password policy, authentication request limits and security-event retention must be specified in the non-functional requirements.

4 Use Cases

This section describes the interactions required to access the application and perform its business operations.

Account holders are distinguished from worker records. An owner authenticates to manage their business. Field Manager access remains conditional on confirmation of that role. Workers do not require application accounts.

UC-04 and UC-20 remain subject to the scope decisions identified in their descriptions.

4.1 Scope and Review Status

The use cases describe the proposed behaviour of OliveTrack, based on the current requirements review.

They remain subject to team review and client validation. Open business decisions are explicitly identified.

The Field Manager is a conditional role. Its inclusion has not yet been confirmed. Whenever included, it operates only within the permissions granted by the owner.

4.2 Shared Definitions

Business context. Each owner has one business workspace containing their properties, campaigns, worker records, documents and operational information.

A business workspace may contain several properties. It is not equivalent to an individual property or campaign.

Information belonging to different business workspaces is not automatically shared or merged.

If the Field Manager role is included, access to an owner's workspace is granted through the applicable invitation and permission rules.

Property and parcel. A property may have optional named parcels. Operational information may identify the whole property or a particular parcel. Unallocated property-level information is not automatically distributed among parcels.

Campaign. A harvest campaign may cover several properties, including whole properties or selected parcels.

Working date. A work period belongs to the date on which it starts, even when it ends after midnight. Full start and end timestamps are used for duration and overlap validation.

Daily attendance. Attendance is recorded once per worker and working date within the business. Actual work assignments identify the relevant campaign and work area.

Confirmation and synchronisation. User confirmation is distinct from server acceptance. A locally confirmed operation may still be pending synchronisation or require resolution.

Current values. Worker-profile changes, work corrections, ticket corrections, analysis-entry corrections and monetary corrections do not require a permanent field-change history.

Distinct records. Document versions, individual signatures and separate source analyses are business records, not merely field-change history. Their preservation follows their specific rules.

Offline preparation. Offline operations use previously prepared, authorised reference information. A two-week download window does not define the permitted duration of an offline session.

Authorisation. Authentication, business membership and responsibility scope are checked separately from permission for the operation. The server checks current authority before accepting protected operations. An access denial must not disclose protected information. The Field Manager participates only if that role is approved and the owner has granted the relevant scope and permission.

Cancellation and validation. Cancelling before confirmation discards the unsaved proposal and preserves accepted records. Invalid input is identified and recoverable information is retained for correction. Case-specific withdrawal, deletion and post-confirmation recovery follow their own explicitly described flows.

Saving and retry. A failed or uncertain transmission does not establish server acceptance or rejection. The interface reports the known storage state, preserves recoverable input and checks or resumes the same operation. A retry must not duplicate its business effect.

Offline submission. Only explicitly supported operations use prepared authorised data. Local confirmation remains pending until server acceptance. Reconnection revalidates current permissions and business rules. Rejected or conflicting proposals remain distinguishable from accepted records and require the applicable recovery workflow.

Financial information. Consultation, registration and correction of monetary amounts and monetary adjustments require online access and financial permission. Offline datasets, summaries and reports exclude them. Mixed-content attachment handling remains pending under Q-OFF-02.

Task and capability. Machinery-operation capability describes a worker profile. Machinery operation and general work describe actual or planned tasks. An unknown actual task remains unclassified; a valid period may still contribute to the counted-hours total.

Provenance and collection. Provenance identifies a source property or parcel. Collection origin distinguishes olives collected from trees or from the ground. Harvesting method identifies how the olives were harvested. These attributes and olive variety are distinct.

Completion and correction. The single definition in Section 6.3.2 governs initial completion, changes to confirmed work and additional periods in closed campaigns.

4.3 Use Case Catalogue

UC-01 Register as a Farm Owner.

UC-02 Log In and Log Out.

UC-03 Recover Account Access.

UC-04 Manage Manager Access and Permissions — conditional on confirmation of the Field Manager role.

UC-05 Manage Properties and Parcels.

UC-06 Manage Harvest Campaigns.

UC-07 Manage Worker Records.

UC-08 Plan Work.

UC-09 Consult Work Schedules.

UC-10 Record Attendance and Actual Work Periods.

UC-11 Correct Work Records.

UC-12 Consult Worker Documents.

UC-13 Upload and Update Worker Documents.

UC-14 Review and Sign a Document.

UC-15 Record Transport and Olive Deliveries.

UC-16 Record Weighing Data and Adjustments.

UC-17 Record Analysis Results.

UC-18 Manage Cooperative-Reported Ticket Values.

UC-19 Consult Campaign Indicators.

UC-20 Generate Worker Documents — scope awaiting confirmation.

4.4 UC-01 — Register as a Farm Owner

Purpose
Allow a new owner to create an account and establish an independent business workspace.

Primary Actor
Visitor.

Preconditions
An internet connection is available. The visitor is not authenticated in the current session.

Required Information
Name of the owner, email address, password and business name.

Main Success Scenario
1. The visitor opens the registration form.
2. The system requests the required information.
3. The visitor enters the information and submits the registration request.

4. The system validates the submitted information and checks whether the email can be registered.
5. The system creates a pending registration and sends an email verification link.
6. The visitor follows the verification link.
7. The system validates the verification request.
8. The system activates the account and creates its business workspace.
9. The system assigns the owner role for that workspace to the new account.
10. The system confirms activation and provides access to the login process.

Alternative and Exception Flows A1. **Invalid information:** The system identifies the fields requiring correction. Registration is not completed.

A2. **Email unavailable for registration:** The system does not create a duplicate account and provides a route to login or account recovery without unnecessarily disclosing account information.

A3. **Verification link expired or invalid:** The system does not activate the account and allows a new verification message to be requested.

A4. **Email delivery failure:** Registration remains pending. The visitor can retry delivery without creating a second account.

A5. **Repeated activation request:** An already processed verification request does not create another account or business workspace.

A6. **Connection failure:** The system reports the incomplete operation and supports a retry without duplicate creation.

Postconditions on Success

A verified account exists. A business workspace exists. The account is its owner. No operational properties, campaigns or workers are created automatically.

Postconditions on Failure

No active business access is granted. A pending registration may remain available for verification or retry.

Connectivity

Registration and email verification require an internet connection.

Related Requirements and Rules FR-001

BR-001, BR-002.

4.5 UC-02 — Log In and Log Out

Purpose

Authenticate an account holder, establish access to an authorised business workspace and allow the user to terminate the session.

Primary Actors

Farm Owner. Field Manager, if this role is included in the approved scope.

Preconditions

For login, an activated account exists and an internet connection is available. For logout, a session exists on the device.

Main Success Scenario — Login

1. The user opens the login screen.
2. The user enters an email address and password.
3. The system validates the credentials and account status.
4. The system determines the user's current business access and permissions.
5. The system establishes an authenticated session.
6. The system opens the authorised workspace.
7. The system makes offline preparation available for the operations supported by the application.

Main Success Scenario — Logout

1. The user selects logout.
2. The system identifies any local changes that have not yet been accepted by the server.
3. The system informs the user of pending changes and provides a synchronisation option when online.
4. The user confirms logout.
5. The system terminates the session on the device and blocks access to the workspace.
6. The system preserves pending changes in protected local storage for recovery through the originating account, subject to renewed authorisation.

Alternative and Exception Flows

A1. **Invalid credentials:** Access is refused using a response that does not unnecessarily reveal which credential was incorrect.

A2. **Unverified account:** Workspace access is refused and the user is offered a route to complete verification.

A3. **No authorised business access:** Authentication does not grant access to a business from which the account has been removed.

A4. **No connection during login:** A new online authentication cannot be completed. Previously authorised offline use follows the established offline-session policy.

A5. **Expired or invalid session:** The system requests renewed authentication before protected online operations continue.

A6. **Logout while offline:** The local session is terminated and workspace access is blocked. Any server-side invalidation requiring connectivity is completed when communication becomes possible.

A7. **Pending synchronisation failure:** The user may still log out. Pending changes are not silently discarded or exposed to another account.

Postconditions on Login

The user can access only the authorised business information and operations.

Postconditions on Logout

The workspace is inaccessible through the terminated local session. Pending data, if any, remains protected and is not transferred to another account.

Connectivity

Login and online permission validation require internet. Supported offline work requires previous authentication and locally available authorised data. Logout is available online and offline.

Related Requirements and Rules FR-002, FR-003, FR-004

BR-002, BR-003, BR-004.

Decision References Q-AA-04, Q-AA-05.

4.6 UC-03 — Recover Account Access

Purpose
Allow an account holder who has forgotten the password to recover access without changing business ownership or permissions.

Primary Actor
Account holder.

Preconditions
An internet connection is available. The account holder can access the registered email inbox.

Main Success Scenario
1. The account holder opens the recovery form.
2. The account holder enters the account email.
3. The system displays a generic acknowledgement.
4. For an eligible account, the system sends a password reset link to the registered email.
5. The account holder follows the link.
6. The system validates the reset request.
7. The account holder enters and confirms a new password.
8. The system validates and saves the new password.
9. The system consumes the reset authorisation and invalidates existing online sessions.
10. The system confirms the change and directs the account holder to login.

Alternative and Exception Flows
A1. **Unknown or ineligible email:** The system displays the same generic acknowledgement without disclosing whether an account exists.

A2. **Invalid, expired or previously used link:** The password is not changed. A new recovery request may be submitted.

A3. **Invalid password or confirmation mismatch:** The system requests correction before completing the reset.

A4. **Email delivery failure:** The existing credentials remain unchanged. The account holder may submit another request.

A5. **Connection failure:** The system reports that completion could not be confirmed and requires validation before reporting a successful reset.

A6. **No access to the registered inbox:** Automatic recovery cannot be completed. An alternative identity-verification process is outside the currently defined workflow.

Postconditions on Success
The new password is active. The reset authorisation cannot be reused. Existing online sessions are invalidated. Business associations and permissions remain unchanged.

Postconditions on Failure
No password change is confirmed. Existing account permissions are unchanged.

Connectivity
Recovery requires an internet connection. Disconnected devices detect session invalidation when they reconnect or reach the offline authorisation limit.

Related Requirements and Rules FR-005

BR-002.

Decision References Q-AA-05, Q-AA-07.

4.7 UC-04 — Manage Manager Access and Permissions

Status

Proposed. Inclusion depends on confirmation of the Field Manager role by the project team and business stakeholders.

Purpose

Allow an owner to invite a Field Manager, define the authorised scope and permissions, and later modify or revoke that access.

Primary Actor

Farm Owner.

Supporting Actor

Invited Field Manager.

Preconditions

The owner is authenticated. An internet connection is available. The owner controls the selected business workspace.

Main Success Scenario — Invite a Manager

1. The owner selects the invitation option.
2. The owner enters the manager's email address.
3. The owner selects the responsibility scope and permitted operations.
4. The system checks the permission configuration.
5. The system creates an invitation and sends an acceptance link.
6. The recipient follows the invitation link.
7. An existing account holder authenticates; a new recipient creates and verifies an account.
8. The system confirms that the authenticated email matches the invitation.
9. The recipient explicitly accepts the invitation.
10. The system activates the manager's access to the owner's business.

Main Success Scenario — Change Permissions

1. The owner selects an existing manager.
2. The system presents the current scope and permissions.
3. The owner changes the configuration.
4. The system validates and saves the changes.
5. Subsequent online requests use the updated authorisation.

Main Success Scenario — Revoke Access

1. The owner selects an existing manager.
2. The owner requests revocation.
3. The system requests confirmation.
4. The owner confirms.

5. The system revokes the manager's business access.
6. Subsequent online requests to that business are refused.

Alternative and Exception Flows A1. **Invitation already pending:** The system allows the owner to review, resend or cancel the existing invitation without creating duplicate memberships.

A2. **Manager already has access:** The system directs the owner to the existing permission configuration.

A3. **Invalid, expired or cancelled invitation:** Access is not granted. The owner may issue a new invitation.

A4. **Different authenticated email:** The invitation cannot be accepted by that account.

A5. **Incomplete mandatory permissions:** The system identifies the incompatible configuration. A manager responsible for workers must retain access to their insurance documents within that scope.

A6. **Invitation cancelled before acceptance:** The invitation no longer grants access.

A7. **Manager device offline during revocation:** Central access is revoked immediately. The disconnected device enforces the change upon reconnection or expiry of offline authorisation.

A8. **Pending changes after revocation:** The revoked manager cannot submit new authorised business changes merely because data was previously available offline.

Existing pending operational proposals may be presented to the owner through an authenticated recovery submission that does not restore the manager's business access.

The owner may review, accept or reject each proposal.

Acceptance requires validation of campaign status, permissions, dates, coverage, dependencies and conflicts.

The record distinguishes the account that originally submitted the information from the owner who authorised its acceptance.

Postconditions on Success

The invitation, permission change or revocation is saved. Access is limited to the owner's business and the authorised responsibility scope.

Postconditions on Revocation

The account remains distinct from its business membership. Existing operational records are preserved. Revocation does not delete worker records or historical business information.

Connectivity

Invitations, acceptance and permission changes require an internet connection.

Related Requirements and Rules FR-004, FR-006, FR-007, FR-008

BR-001, BR-002, BR-003, BR-004.

Decision References Q-AA-01, Q-AA-02, D-01, D-08.

4.8 UC-05 — Manage Properties and Parcels

Purpose Allow an authorised user to create, consult, update, archive, reactivate and, where permitted, delete properties and their optional parcels.

The current agricultural scope concerns olive harvesting.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, when explicitly granted property-management permissions by the owner.

Permission to record operational work does not automatically grant permission to manage properties.

Preconditions

- The user has permission for the requested operation.
- Creation, modification, archival, reactivation and deletion require an internet connection.
- For operations on an existing record, that record exists within the authorised scope.
- To create a parcel, its property exists.
- Offline consultation requires previously downloaded authorised information.

Main Flow

1. The user opens property management.
2. The user chooses to create a property or selects an existing property.
3. The system presents the existing information or an entry form.
4. The user enters or updates the property name and textual location.
5. The user may enter the declared area in hectares and one or more olive varieties.
6. The system identifies the owner associated with the property.
7. If internal divisions are needed, the user creates or updates named parcels. Parcel area and olive varieties are optional.
8. The system validates the information and access permissions.
9. The user reviews and confirms the changes.
10. The system saves the records and confirms the result.

For consultation only, the flow ends after the requested information is displayed.

Alternative Flows and Exceptions

A1 — No internal divisions are needed. The user saves the property without parcels. The absence of parcels does not prevent use of the property in operational records.

A2 — Optional information is unknown. The user leaves area or variety information unfilled and may add it later. An unknown area is not treated as zero.

A3 — The user archives a record. The system requests confirmation and archives the selected property or parcel. Historical records remain accessible within the user's permissions. The archived record is unavailable for new operational associations.

A4 — The user reactivates a record. The system validates the request and restores its availability, subject to its parent property's state.

A5 — The user requests deletion. The system checks for associated records. If none exist, the user may confirm deletion. Otherwise, deletion is refused and archival is offered.

A6 — Connectivity is unavailable. The system permits consultation of previously downloaded authorised records and identifies their synchronisation status. Management operations are unavailable until connectivity is restored.

Postconditions **On successful creation or update:** Property and parcel information is saved with the correct ownership and parent associations.

On archival: Historical information is preserved, and new operational use is blocked for the archived scope.

On reactivation: The record becomes available again, subject to the property's availability.

On deletion: Only the eligible record without associated records is removed.

On consultation, cancellation or rejection: Stored information remains unchanged.

Related Requirements and Rules FR-009, FR-010, FR-011, FR-012, FR-013

BR-005, BR-006, BR-013.

Decision References Q-SCOPE-01, D-06.

4.9 UC-06 — Manage Harvest Campaigns

Purpose Allow an authorised user to create, consult, update, close, reopen, archive, unarchive and, where permitted, delete harvest campaigns.

A campaign may cover one or more properties, including whole properties or selected parcels.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, with the relevant permissions granted by the owner.

Permissions apply to the requested operation. Operational access does not automatically grant all campaign-management permissions.

Preconditions

- The user has permission for the requested operation.
- Campaign-management operations require an internet connection.
- Selected properties and parcels exist within the authorised scope.

- For operations on an existing campaign, that campaign exists.
- Offline consultation requires previously downloaded authorised information.

Main Flow

1. The user opens campaign management.
2. The user chooses to create a campaign or selects an existing one.
3. The system presents its information or an entry form.
4. The user enters or updates the campaign name and planned start date.
5. The user selects one or more properties.
6. For each property, the user chooses the whole property or specific parcels.
7. The system validates the selected areas, dates and permissions.
8. The system checks for conflicting campaign coverage.
9. The user reviews and confirms the proposed information.
10. The system saves the campaign and confirms the result.

For consultation only, the flow ends after the authorised campaign information, dates and covered areas are displayed.

Alternative Flows and Exceptions

A1 — Parcels do not yet exist.
The user may create parcels through UC-05, provided they have the required permission. The parcels remain associated with the property for future use. The user then returns to campaign selection.

A2 — Conflicting campaign coverage.
An open campaign blocks another campaign from covering the same area, including a campaign with a future planned start date.

Whole-property coverage conflicts with any parcel of that property. Campaigns covering different parcels may coexist.

The system also checks historical overlap against closed campaigns using their applicable recorded dates.

Archiving a campaign does not bypass these checks.

A3 — Areas are added.
The system checks permissions and campaign conflicts before accepting the additional coverage.

A4 — Areas are removed or an incorrect coverage association is resolved.
An area without operational records may be removed after permission and coverage validation.

If operational records exist, the system identifies the dependent information.

For a closed campaign, the user must first obtain authorised reopening before changing coverage or reassigning operational records.

The user resolves the dependencies through the applicable correction or permitted deletion workflows.

Only after all dependent records have been resolved may the area be removed.

Removing coverage does not automatically delete work, deliveries, tickets, analyses or reported amounts. Unresolved dependencies block removal.

A5 — The user closes the campaign.
The user provides or confirms the actual end date. The system validates the closure against existing activity and the agreed date rules. On confirmation, the campaign is closed.

A6 — External information arrives after closure.
An authorised user may add or complete ticket information, analysis results and cooperative-reported values for existing deliveries and tickets. This does not require reopening the campaign.

A7 — Missing earlier work is completed.
An authorised user may add previously unrecorded work information to a closed campaign when the work occurred within its valid actual dates.

This may include completing missing hours or tasks and adding a previously unrecorded actual work period.

The operation must not replace or remove confirmed values, change existing confirmed associations or create contradictory attendance or overlapping periods.

Unknown information remains explicitly incomplete.

A8 — Confirmed work requires correction.
Changing or removing previously confirmed attendance, actual assignments, periods, tasks, breaks or associations requires authorised campaign reopening.

A completion operation must not be used to bypass the correction workflow.

A9 — Work occurred after the end date.
The campaign must be reopened and its dates reviewed before that work is accepted.

A10 — The user reopens the campaign.
The system checks reopening permission and conflicting campaign coverage. Existing records are preserved. Previous closure information must not be mistaken for the campaign's current lifecycle status.

A11 — The user archives or unarchives.
The system changes visibility in the usual lists without changing the closure status. Archiving does not close a campaign. Unarchiving does not reopen one. Archiving must not bypass campaign-overlap validation.

A12 — The user requests deletion.
The system checks for operational records. If none exist, deletion may be confirmed. Otherwise, deletion is refused and archival is offered.

A13 — Connectivity is unavailable.
Management operations cannot be completed. Previously downloaded authorised campaign information remains available for consultation and supported field operations.

A14 — The actual start is recorded manually.
An authorised user records and confirms the actual start date.

The system validates the date against the campaign context and existing operational information.

The planned start remains unchanged unless separately edited.

A15 — Actual work provides the first start event.
When the first actual campaign-area assignment is confirmed and no actual start exists, the system proposes its working date as the campaign's actual start.

The user confirms or corrects the proposal.

Recording attendance without an actual campaign assignment does not trigger this proposal.

If the device is offline, operational work may be saved locally, but the campaign-management change requires connectivity and server confirmation.

Postconditions **On successful creation or update:** The campaign is saved with its name, planned start date and selected coverage.

On closure: The campaign has an actual end date. Later completion of relevant external information remains possible without reopening. Initial completion and additional periods follow Section 6.3.2. Work-record corrections require authorised reopening.

On reopening: The campaign permits further work within its revised valid scope and dates.

On archival or unarchival: Visibility changes without deleting history or changing the closure status.

On deletion: The eligible campaign is removed without deleting operational history.

On consultation, cancellation or rejection: Stored information remains unchanged.

Related Requirements and Rules FR-010, FR-014, FR-015, FR-016, FR-017, FR-018, FR-019

BR-006, BR-007, BR-008, BR-009, BR-010, BR-011, BR-012, BR-013.

Decision References D-02, D-03, D-06, D-04.

4.10 UC-07 — Manage Worker Records

Purpose Allow an authorised user to create, consult, update, inactivate, reactivate and, where permitted, delete worker records within a business.

Worker records support operational management and document workflows. They do not constitute worker application accounts.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if this role is included and the owner grants the relevant permissions.

Permission to access worker information does not automatically grant access to every associated document.

Preconditions

- The user has permission for the requested operation.

- Management operations require an internet connection.
- For operations on an existing worker, the record exists within the authorised business context.
- Offline consultation requires previously downloaded authorised information.

Main Flow

1. The user opens worker management.
2. The user searches for an existing worker or chooses to create a record.
3. The system displays the authorised information or an entry form.
4. The user enters or updates the worker's full name and telephone number.
5. The user indicates whether the worker can operate machinery and may provide an optional description.
6. The system validates required fields and checks for matching names or telephone numbers within the same business.
7. The user reviews and confirms the information.
8. The system saves the current values and updates the last-modified date.
9. The system confirms the result.

For consultation only, the flow ends after the authorised worker information is displayed.

Alternative Flows and Exceptions

A1 — No worker matches the search. The system displays an explicit no-results message and permits authorised creation of a new record.

A2 — A matching name or telephone is found. The system warns of a possible duplicate within the same business. The user may open the existing record or confirm that a separate record is needed. Records are not automatically merged.

A3 — The user inactivates a worker. The system requests confirmation and removes the worker from the usual selections for new work. Historical work and documents are preserved.

A4 — The user reactivates a worker. The system restores the worker's availability for selection without creating another identity.

A5 — The user requests deletion. The system checks for associated records. Deletion may be confirmed only if none exist. Otherwise, deletion is refused and inactivation is offered.

A6 — A creation request is repeated. Retrying the same submission does not unintentionally create duplicate records.

A7 — Connectivity is unavailable. The system permits consultation of previously downloaded authorised information. Creating, editing, inactivating, reactivating or deleting records requires reconnection.

Postconditions **On successful creation or update:** The current worker information is saved within the correct business context, with its last-modified date.

On inactivation or reactivation: Selection availability changes without deleting historical records.

On deletion: Only an eligible record without associations is removed.

On consultation, cancellation or rejection: Previously saved information remains unchanged.

In all cases: The operation does not automatically create an application account, schedule work or confirm attendance.

Related Requirements and Rules FR-020, FR-021, FR-022, FR-023

BR-001, BR-014, BR-015, BR-016, BR-031.

Decision References Q-DOC-02.

4.11 UC-08 — Plan Work

Purpose Allow an authorised user to plan workers, work areas, optional periods and tasks for one or more working days.

Planning may start from a new entry, an existing plan or previously recorded actual work. It never establishes attendance or actual hours for the target dates.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if this role is included and the owner grants the relevant planning permissions.

Preconditions

- The user has permission to plan work within the selected scope.
- The campaign, work areas and worker records exist.
- Selected areas are included in the campaign coverage.
- The requested dates comply with the campaign lifecycle rules.
- Offline planning requires the necessary authorised reference data to be available on the device.

Main Flow

1. The user selects the campaign and one or more target dates.
2. The user selects a property or an included parcel.
3. The system displays existing plans for the selected context.
4. The user selects the workers expected to work.

5. The user may enter planned periods, task types and optional breaks. Common values can be applied to a group and adjusted individually.
6. The system checks required information, worker availability, area eligibility and known scheduling conflicts.
7. Where a worker is planned for several areas on the same day, the system requires periods for the relevant assignments.
8. The system presents the proposed daily plans and any unresolved issues.
9. The user reviews and confirms.
10. The system saves independent daily plans and displays their local and server status.

Alternative Flows and Exceptions

A1 — A minimal plan is sufficient. For a worker assigned to one area on a working day, the user may save the campaign, date, area and worker without specifying hours or task type.

A2 — Individual details differ. The user adjusts one worker's periods or tasks without unintentionally changing other workers' entries.

A3 — Several areas apply. The user creates the relevant assignments. Periods are required for all assignments of that worker to different areas on the same working day.

A4 — Several target dates are selected. The system applies the reviewed proposal to each selected date. Each date receives an independent plan, validated against its own context.

A5 — An existing plan is copied. The user selects a source plan and the target dates. Workers, areas, periods and tasks are proposed for review. Source plans remain unchanged.

A6 — Actual work is used as a basis. The user selects previously recorded work. Its workers, areas, periods and tasks are proposed as planning information for the target dates. Confirmed attendance and actual work records are not copied to those dates.

A7 — Target dates already contain plans. The system identifies the existing plans. Reuse does not silently overwrite or duplicate them. Changes require explicit review.

A8 — A worker is inactive. The system identifies the inactive worker and requires an eligible selection or authorised reactivation. Existing future plans are flagged rather than silently edited.

A9 — Periods overlap. The system blocks conflicting periods, considering the worker's relevant plans across properties, parcels and campaigns.

A10 — Work crosses midnight. The user records the start and end with sufficient date information. The period belongs to the working date on which it starts. Validation considers the full interval, including its continuation into the next day.

A11 — A worker is removed or replaced. Before actual work has been recorded for the affected daily area context, the user may confirm removal or replacement. Other assignments remain unchanged.

A12 — Actual work already exists. The corresponding plan remains the reference for what was expected. Differences are recorded as actual work or handled through the work-correction process.

A13 — The device is offline. The user may create or adjust plans using prepared authorised data. The system identifies the changes as pending synchronisation.

A14 — Planned breaks are entered. The user identifies break start and end times and their intended counting treatment. The system prevents duplicate deductions in any displayed planned-duration calculation. These values remain subject to actual confirmation.

Postconditions **On success:** Daily plans identify the selected campaign, working dates, areas and workers, with any supplied periods and tasks.

On reuse: Target plans are independent of their source. Editing a target day does not modify the source or other target days.

On pending synchronisation: Local plans remain distinguishable from server-accepted plans.

On cancellation or rejection: Previously accepted information remains unchanged.

In all cases: Planning does not confirm attendance, create actual hours or modify source actual work.

Related Requirements and Rules FR-024, FR-025, FR-026, FR-027, FR-029

BR-015, BR-016, BR-017, BR-019, BR-021.

Decision References Q-WF-04, Q-AA-04, Q-WF-02.

4.12 UC-09 — Consult Work Schedules

Purpose Allow an authorised user to consult planned work by day, by worker or by week, within the selected harvest campaign.

Consultation does not modify plans or establish actual attendance.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if this role is included and the owner grants the relevant consultation permissions.

Preconditions

- The selected campaign exists.
- The user is authorised to consult the requested information.
- Offline consultation requires the relevant plans and reference information to be available on the device.

An existing plan is not a precondition: a valid query may return no planned work.

Main Flow

1. The user opens schedule consultation and selects a campaign.

2. The user chooses a daily, worker-based or weekly view.
3. The user selects the relevant date, week or worker consultation period.
4. The user may filter by property, parcel, worker and planned task type.
5. The system retrieves the matching authorised plans.
6. The system displays the workers, working dates, properties or parcels, planned periods, task types and any recorded planned breaks.
7. Missing hours or task types are identified as not yet defined.
8. The system identifies relevant synchronisation and freshness information.
9. The user may change the view or filters and repeat the consultation.

Alternative Flows and Exceptions

A1 — No matching plan exists. After a successful query, the system states that no planned work matches the selection. This does not establish worker absence.

A2 — A plan has incomplete details. The system displays the known information. Missing hours or task types are shown as not yet defined, not as zero hours or an assumed task.

A3 — Several areas are included. The system can display all authorised areas of the selected campaign together. Each assignment retains its property and optional parcel identification.

A4 — A period crosses midnight. The system displays the period under its start date and identifies the end date and time.

A5 — A planned worker is inactive. The system identifies the inactive status. The saved assignment is not silently removed. Any change follows UC-08.

A6 — Local changes are pending. The system identifies locally saved plans or changes awaiting synchronisation. They are not presented as server-confirmed.

A7 — The device is offline. The system displays prepared authorised plans for the current and following week, together with the last successful synchronisation information.

A8 — Requested information is not local. The system explains that the information is unavailable without connectivity. It does not claim that no plan exists.

A9 — Retrieval fails. The system explains the failure and offers a retry where appropriate. A failed query is distinguishable from a successful query with no results.

A10 — Access is restricted. The system excludes unauthorised information and identifies the scope being displayed without exposing protected details.

Postconditions **On success:** Matching authorised plans are displayed, or a successful query confirms no matching plans.

On offline or incomplete consultation: The system identifies the available scope and relevant limitations.

In all cases: Consultation does not modify plans, confirm attendance or create actual hours.

Related Requirements and Rules FR-028

BR-017, BR-019, BR-021.

Decision References Q-AA-04, Q-AA-05.

4.13 UC-10 — Record Attendance and Actual Work Periods

Purpose Allow an authorised user to confirm daily attendance and record actual work periods, tasks, locations and optional breaks.

Planned information may support entry, but becomes actual work only after explicit confirmation.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if this role is included and the owner grants the relevant permissions.

Workers are not required to use an application account to confirm their own attendance.

Preconditions

- The user has permission for the requested operation.
- Worker records already exist.
- The relevant campaign and work areas exist within the authorised scope.
- Work dates comply with the campaign lifecycle rules.
- Offline entry requires the necessary authorised reference data to be available.

Main Flow

1. The user selects the campaign and working date.
2. The system displays planned workers, available proposed periods and any existing attendance and work records.
3. Workers without an attendance decision are shown as awaiting confirmation.
4. The user marks workers as present or absent. Daily attendance is shared within the business for that worker and working date.
5. For present workers, the user identifies the actual campaign and work-area assignments. These may remain incomplete as to hours or tasks.
6. The user reviews proposed periods or enters actual periods, tasks and areas. Common values may be applied to a group with individual adjustments.
7. The user reviews or records actual breaks, including their start, end and whether they count towards working time. Planned breaks are not accepted automatically.
8. The system validates the entries and calculates available working-time totals.
9. The user reviews and explicitly confirms the submitted information.
10. The system saves the records and displays their local storage and synchronisation status.

Alternative Flows and Exceptions

A1 — Attendance is known before hours. The user saves presence or absence and completes work details later. Missing hours remain incomplete, not a confirmed zero total.

A2 — An unplanned worker attends. The user selects an existing authorised worker record and records attendance. A planned assignment is not required.

A3 — No plan exists. The user selects existing workers and enters the actual information directly.

A4 — Actual work differs from the plan. The user adjusts workers, periods, tasks or areas before confirmation. The original plan remains unchanged. A valid period with an unknown task contributes to total counted hours as unclassified, separately from named task totals.

A5 — A worker changes task or area. The user records separate periods for the relevant campaign and area. Known tasks are identified; an unknown task remains unclassified. Valid unclassified hours contribute to total counted hours.

A6 — Work periods exclude a break. The interval between those periods does not contribute to working time. The same excluded interval is not deducted again.

A7 — A break occurs within a period. The user records its start and end. It is deducted only when explicitly marked as not counted towards working time.

A8 — A worker is absent. The system does not convert planned periods into actual work. Contradictory actual work entries require resolution.

A9 — Work crosses midnight. The period retains its start-date association. Its end date and time establish the full interval for calculations and overlap validation.

A10 — Work is entered retrospectively. The user selects the actual working date. The system preserves the distinction between work date and entry date.

A11 — The campaign is closed. The user may complete previously missing information for work performed within the campaign's valid actual dates.

This includes entering missing details or a previously unrecorded actual period, provided that confirmed values and associations remain unchanged.

The system validates attendance consistency, permissions, campaign coverage and time overlaps.

Changing or removing confirmed information requires authorised reopening and follows UC-11.

Work outside the campaign's valid actual dates requires reopening and date review.

A12 — Confirmed information needs correction. The user follows UC-11. Existing values are not silently replaced through another initial submission.

A13 — Invalid or overlapping periods. The system identifies the problem and preserves unaffected input. Validation considers the worker's relevant periods across areas and campaigns.

A14 — The device is offline. The system saves supported entries locally. Their synchronisation status remains visible.

Recording actual work does not silently change campaign dates.

If the campaign has no actual start, the first confirmed actual assignment provides a proposed start date under UC-06.

An offline assignment may remain saved locally while the campaign date awaits connected confirmation.

Postconditions **On success:** The submitted attendance and available actual work details are saved in the correct context.

On attendance-only confirmation: Presence is recorded while hours and tasks remain explicitly incomplete where unknown.

On offline save: User-confirmed entries are stored locally and remain pending server acknowledgement.

On cancellation or rejected changes: Previously saved information remains unchanged.

In all cases: Actual work entry does not rewrite the plan.

Related Requirements and Rules FR-016, FR-030, FR-031, FR-032, FR-033, FR-034, FR-035

BR-011, BR-018, BR-019, BR-020, BR-021.

Decision References Q-WF-01, Q-WF-03, D-04.

4.14 UC-11 — Correct Work Records

Purpose Allow an authorised user to correct saved attendance and actual work information, including assignments without hours, periods, tasks, work areas and breaks.

Corrections update the affected totals without rewriting the original work plan.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if included and granted explicit correction permissions by the owner.

Permission to record work does not automatically grant permission to correct confirmed records.

Preconditions

- The target record exists.
- The user has correction permission for the affected records and contexts.
- Campaigns whose work records are being changed are open.
- Offline correction requires the relevant authorised data to be available on the device.

Main Flow

1. The user selects the worker, working date and relevant context.
2. The system displays current attendance, periods, tasks, areas, breaks and totals.

3. The user proposes the correction.
4. The system validates permissions, campaign status, dates, associations, time intervals and break treatment.
5. The system checks the worker's relevant periods for overlaps.
6. The system presents the corrected information and affected totals for review.
7. The user confirms.
8. The system saves the current values and updates the affected local totals.
9. The system displays whether the change is pending synchronisation or accepted by the server.

Alternative Flows and Exceptions

A1 — A campaign is closed. Correction cannot proceed until an authorised user reopens the campaign through UC-06. No additional correction deadline applies.

A2 — Presence is changed to absence. The system identifies all actual campaign-area assignments, including those without hours, and all periods and breaks associated with that worker and working date within the business.

The user must have the required permissions, and affected campaigns must satisfy the reopening rules.

The system requests confirmation before removing the affected work and changing attendance. It must not leave the worker absent while contradictory confirmed assignments, periods or breaks remain.

A3 — Record associations are incorrect. The user may correct the worker, working date, campaign, property or parcel.

The system validates the origin and destination contexts, campaign status and time overlaps. It also checks daily-attendance consistency.

Only the selected work records are reassigned. Other periods belonging to the original worker remain unchanged.

Affected origin and destination totals are updated after acceptance.

A4 — Periods or breaks are invalid. The system identifies the problem and preserves unaffected input. Corrections must not create overlapping actual periods or duplicate break deductions.

A5 — The device is offline. The system saves the correction locally and identifies it as pending. Local totals reflect the pending change. Server permissions, campaign status and consistency are checked on reconnection.

A6 — The server record has changed. The system presents the current server values and the pending correction to an authorised user for manual resolution. No conflicting update is silently substituted.

A7 — The user resolves a conflict. The user may retain the server values or review and confirm a revised correction. Any submitted correction is validated again.

Postconditions **On local acceptance:** The corrected values and affected local totals are saved with their synchronisation status.

On server acceptance: Central records and dependent totals reflect the accepted correction. Other clients receive the change through synchronisation.

On cancellation or rejection: Authoritative accepted records remain unchanged. Any unresolved local proposal is clearly identified.

In all cases: Original plans and unrelated records are not modified.

Related Requirements and Rules FR-033, FR-036, FR-037, FR-061

BR-011, BR-018, BR-019, BR-020, BR-021.

Decision References Q-WF-04, D-04.

4.15 UC-12 — Consult Worker Documents

Purpose Allow an authorised user to find, open and download worker documents, including uploaded files.

Uploaded documents are included in the current scope.

Application-generated documents may appear in the same consultation workflow if generation is confirmed through UC-20.

Insurance documents must support prepared offline consultation for field operations.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if this role is included.

The owner may consult documents within their business.

A Field Manager must have access to insurance documents for workers within their authorised operational scope. Contracts and other documents require additional owner-granted permissions.

Preconditions

- The worker record exists.
- The user is authorised to consult the requested document type and worker scope.
- Offline access requires the relevant authorised files and metadata to have been prepared on the device.

An existing document is not a precondition: the document list may be empty.

Main Flow

1. The user selects a worker and opens the document list.
2. The system displays authorised documents, organised as contracts, insurance or other documents.

3. The system identifies each document by its title, type, relevant date or version and signature status where applicable.
4. Insurance entries display the recorded expiry date and validity status.
5. The user selects an available version.
6. The system verifies access and opens the PDF, JPG or PNG file.
7. The user may download the authorised document.

Alternative Flows and Exceptions

A1 — No documents are available.
The system states that no documents are available within the authorised scope.

A2 — A previous version is requested.
For contracts, all retained versions are consultable.

For insurance and other documents, only the current and immediately previous versions are retained and consultable.

Older insurance and other-document versions are deleted under UC-13, including versions that were signed.

A3 — Insurance validity is incomplete.
If the expiry date is missing, the system displays validity as unconfirmed. It does not infer validity from upload date or version order.

A4 — Insurance is expired.
The system identifies the expired status while allowing authorised consultation. Expiry does not hide the document.

A5 — The device is offline.
The system opens prepared authorised insurance files and displays the last successful synchronisation information. The current and previous versions are available where they exist and have been successfully prepared.

A6 — A required file is not local.
The system explains that the document is unavailable offline. It does not claim that no document exists.

A7 — Additional workers need preparation.
While connected, the authorised user may prepare insurance documents for additional existing workers who may be called to work.

A8 — Retrieval or download fails.
The system explains the failure and offers a retry where appropriate. Failure is distinguishable from an empty document list.

A9 — A shared document is opened.
The document may be associated with several workers without duplicate file storage.

Within the selected worker's view, application-captured signature information is limited to that worker.

A10 — A document was signed outside the application.
The interface identifies the external-signature declaration separately from application signing. It does not claim that the system verified the external signature.

A11 — A retained version is withdrawn.
The interface identifies it as withdrawn and does not present it as available for operational use. Its retained signature information is not changed to unsigned.

A12 — A deleted version remains on an offline device.
The device applies the authorised deletion when it reconnects and synchronises. The interface must not promise knowledge of server changes that have not yet been received.

Postconditions **On success:** The selected authorised document version is displayed or downloaded.

On offline consultation: The available local version and its freshness are identifiable.

On failure: The system explains the limitation without exposing protected information.

In all cases: Consultation does not change document contents, version identity or signature status.

Related Requirements and Rules FR-038, FR-039, FR-041, FR-044

BR-022, BR-023, BR-024, BR-025, BR-026, BR-027, BR-028, BR-029, BR-030.

Decision References Q-DOC-07, Q-DOC-04, Q-SCOPE-01.

4.16 UC-13 — Upload and Update Worker Documents

Purpose Allow an authorised user to upload documents and create new versions associated with one or more workers.

Document generation and signing are handled through separate use cases.

This use case covers existing files supplied for upload. Document generation remains a separate candidate capability under UC-20.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if included and granted upload or update permissions for the relevant document type.

Insurance consultation permission does not automatically permit document modification.

Preconditions

- An internet connection is available.
- The user has permission for the document type and affected workers.
- Associated worker records exist within the same authorised business.
- For a version update, the target document exists.
- A supported file is available.

Main Flow

1. The user opens document management.
2. The user chooses a new document or a new version of an existing document.
3. The user selects or reviews the associated workers.
4. The user selects the document category: contract, insurance or other.
5. The user provides or confirms the document title and selects the file.
6. For insurance, the user enters the expiry date where known.
7. The user may add observations and declare whether the uploaded version was signed outside the application.
8. The system validates permissions, required fields, file format and size.
9. The system presents the submission and relevant version-retention consequences.
10. The user confirms.
11. The system stores the accepted version, records its upload date and version identity, and makes it current.
12. The system applies the retention policy and confirms the outcome.

Alternative Flows and Exceptions

A1 — An insurance expiry date is unknown. The system allows the document to be saved with validity identified as unconfirmed. It does not infer validity from upload date.

A2 — A document covers several workers. The user associates the same document with the relevant authorised workers. The file is not duplicated for each worker.

A3 — A version was signed externally. The user explicitly identifies that status for the uploaded version. It is distinguishable from an application signing event and is not presented as externally verified by the system.

A4 — An existing signed document is updated. The new version becomes current without inheriting an earlier signature. Earlier versions retain their own status while they remain stored.

A5 — Retention removes an older version. For insurance and other documents, the system retains the current and immediately previous versions only. Older versions are deleted, including versions marked as signed. This policy applies to the shared version, not separately to each worker association.

A6 — An incorrect upload may be deleted. An authorised user may confirm deletion when the target has no signature and no dependent records.

A7 — An incorrect upload is signed. The user may withdraw it from use, preserving its file and signature association while it remains retained. The rolling retention policy for insurance and other documents remains applicable.

If the withdrawn version is current, it remains identified as current and withdrawn until a replacement is accepted.

An earlier version does not automatically become current.

A8 — Validation fails. The system identifies unsupported formats, files exceeding the size limit, missing required information or invalid associations. No version is accepted until corrected.

A9 — Upload or storage fails. The system reports the failure. The previously current version and its retained history remain unchanged. Failed uploads do not trigger retention deletion.

A10 — A submission is repeated. Retrying the same upload operation does not unintentionally publish duplicate versions or trigger extra deletion.

A11 — Connectivity is unavailable. Upload and update operations cannot complete. Prepared offline insurance consultation remains available through UC-12.

Postconditions **On success:** The accepted version becomes current and is available to authorised consultation through the associated worker records.

On insurance or other-document update: Only the current and immediately previous versions remain stored.

On contract update: All accepted versions remain stored, subject to separately authorised document actions.

On failure or cancellation: Previously accepted versions are unchanged and no retention deletion is performed.

In all cases: A new version does not inherit signatures from another version.

Related Requirements and Rules FR-040, FR-041

BR-022, BR-023, BR-024, BR-025, BR-026, BR-027, BR-028, BR-030.

Decision References Q-DOC-06, D-07, Q-DOC-07, Q-DOC-01.

4.17 UC-14 — Review and Sign a Document

Purpose Allow a worker to review a specific document version and confirm a drawn signature in person on an authorised user's device.

The signing result belongs to that worker and the reviewed version.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Initiating actor:** Farm Owner.

Conditional initiating actor: Field Manager, if included and granted the relevant permissions.

Signing actor: Worker.

The worker does not require an application management account.

This process collects the worker's signature only. Any required owner signature is handled separately.

Preconditions

- The worker and document exist.

- The initiating user has permission to access the document and initiate signing.
- The selected version is available and eligible for signing.
- The worker is physically present.
- The worker has not already completed this signing process for the selected version.
- Offline signing requires the authorised document and relevant data to be available on the device.

Main Flow

1. The initiating user selects the worker and document version.
2. The system clearly identifies the intended worker and document.
3. The initiating user confirms that the selected worker is present.
4. The system presents the document for the worker to review.
5. The worker chooses to proceed.
6. The system displays a signature drawing area.
7. The worker draws a signature using a finger or compatible pen.
8. The worker reviews the drawing and its association with the document.
9. The worker explicitly confirms the signing action.
10. The system saves the signature result with the worker, version, initiating user and recorded date and time.
11. The system displays whether the result is pending synchronisation or accepted by the server.

Alternative Flows and Exceptions

A1 — The worker redraws the signature.
Before confirmation, the worker may clear the drawing and try again.

A2 — The worker cancels.
No completed signature is created by the cancelled interaction. The document is not marked as signed for that worker by this interaction.

A3 — No signature has been drawn.
The system requests a drawing before accepting confirmation.

A4 — The document cannot be reviewed.
Signing cannot proceed. The system explains the access or retrieval limitation.

A5 — A signing result already exists.
If the worker has already confirmed a signature for the selected version, the system displays its actual state.

A pending or uncertain result is checked or resumed rather than replaced by another signing interaction.

An accepted signature is not overwritten.

A6 — The document is shared.
Each worker completes an individual review and signing interaction. One worker's signature does not establish another worker's signature.

A7 — The device is offline.
The system saves the confirmed result locally as pending synchronisation. It does not claim server acceptance.

A8 — A newer version exists.
The signature remains linked to the version actually reviewed. It is not transferred to the newer version.

A9 — An older version is synchronised.
If the original version remains available and eligible, the server may accept the signature for that version. The system identifies that a newer version exists.

A10 — The original version is withdrawn or deleted.
The pending result requires explicit resolution.

A withdrawn version does not automatically accept a pending signature.

A version deleted under the retention policy is not recreated merely because an offline signing operation reconnects.

No newer version is marked as signed by that pending operation.

A11 — The submission outcome is uncertain.
The system checks the existing operation before initiating another signing event. It does not automatically ask the worker to sign again.

Postconditions **On local confirmation:** The signature result and identifying information are saved on the device. The result is pending until server acceptance where connectivity is unavailable.

On server acceptance: The signature is associated with the intended worker and reviewed version and is available for authorised consultation.

On cancellation: No completed signature is created by the cancelled interaction.

On unresolved submission: The pending or rejected state is explicit. No other document version is marked as signed.

In all cases: Participation does not grant additional application permissions.

Related Requirements and Rules FR-041, FR-042, FR-043, FR-044, FR-045

BR-023, BR-026, BR-027, BR-028, BR-029, BR-030.

Decision References Q-DOC-02, Q-DOC-04, Q-DOC-07.

4.18 UC-15 — Record Transport and Olive Deliveries

Purpose Allow an authorised user to record and ordinarily correct the departure and delivery details of an olive load, including provenance and transport information.

A delivery may be recorded in two stages or entered retrospectively after arrival.

Weighing tickets, analysis results and monetary values are handled separately.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant permissions.

The driver is identified through an existing worker record and does not need an application account.

Preconditions

- The user is authorised to record the transport or confirm the delivery.
- The campaign and relevant properties or parcels exist within the authorised scope.
- The driver has an existing worker record within the business.
- Selected areas are included in the campaign coverage.
- Offline entry requires the relevant authorised reference data to be available.

Main Flow

1. The user starts a delivery record and selects the campaign.
2. The user identifies the property or properties of origin and may select specific parcels.
3. The user may identify collection origin and olive variety for each recorded part of the load.
4. The user selects the driver from existing worker records.
5. The user identifies the vehicle.
6. The user selects a previously used destination or enters a new destination name.
7. The user records the actual departure date and time.
8. The system validates required information, associations and permissions.
9. The user reviews and confirms.
10. The system saves the delivery as in transport and displays its synchronisation status.
11. After arrival, an authorised user opens the record and confirms delivery, providing the actual delivery date and time.
12. The system validates and saves the delivered status.

Alternative Flows and Exceptions

A1 — The delivery already occurred. The user enters departure and delivery information in the same operation. After validation and confirmation, the record is saved as delivered.

A2 — The load has several provenances. The user records the relevant properties and optional parcels separately. Missing quantities are not invented or automatically allocated.

A3 — Collection origin or variety is unknown. The user may leave the information unfilled. The system does not assume tree-collected or ground-collected olives.

A4 — The destination is new. The user enters its name directly. A separate cooperative-management process is not required.

A5 — The driver has no worker record. The user cannot select an external driver by free-text substitution. A worker record must first be created through UC-07 while connected.

A6 — Weighing information is unavailable. Transport and delivery can be recorded without ticket weights. Missing weights are not treated as zero. Each ticket will belong to this delivery under UC-16. Whether one delivery may have several tickets remains an open business question.

A7 — Analysis or monetary values are unavailable. The delivery remains recorded. Results and reported amounts may be added later through their respective use cases.

A8 — The device is offline. Departure and delivery confirmation can be saved locally using prepared data. A new destination name may be entered. Changes remain pending synchronisation.

A9 — Saved transport or delivery details require correction. An authorised user may review and correct existing details within the same campaign and operational state. The system validates permissions, chronology and affected associations before accepting the correction. Reversing confirmation, cancellation or moving the delivery to another campaign requires a separately approved exceptional procedure under Q-DA-06.

Postconditions **After departure entry:** The delivery has its required transport and provenance information and is identified as in transport.

After delivery confirmation: The record identifies the actual delivery date and time and is marked as delivered.

On offline save: The physical delivery status is distinguishable from its pending synchronisation status.

On cancellation or rejected submission: Previously accepted information remains unchanged.

In all cases: Unknown weights, analysis results and monetary values remain unknown, rather than being replaced with zero.

Related Requirements and Rules FR-046, FR-047, FR-048

BR-001, BR-031, BR-035, BR-036.

Decision References Q-DA-06, Q-DA-07, Q-DA-10.

4.19 UC-16 — Record Weighing Data and Adjustments

Purpose Allow an authorised user to record weighing ticket information and reported adjustments for an olive delivery.

Values are entered manually and retain their source meaning. Incomplete information may be saved.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant permissions.

The cooperative supplies ticket information. Direct integration and automatic ticket reading are outside the current scope.

Preconditions

- The delivery exists.
- The user has permission for the requested entry or correction.
- Source information is available for manual entry.
- Offline entry requires the relevant authorised delivery information on the device and is limited to non-financial data. Monetary information follows FR-053. Mixed-content attachments require the policy pending under Q-OFF-02.

Main Flow

1. The user selects the delivery and opens its weighing information.
2. The user creates a ticket record or opens an existing one.
3. The user enters the available ticket number or reference, cooperative and weighing date and time.
4. The user enters the known weights, identifying their meaning and units.
5. The user records known operational weight adjustments as reported by the cooperative. Monetary adjustment information requires online access and the financial permissions of UC-18.
6. Where known, the user identifies whether an adjustment is already reflected in a reported weight. Monetary effects are handled online as financial information and are not copied into offline datasets.
7. The user may attach an image or PDF of the ticket.
8. The system validates the available information without inventing missing values.
9. The system presents source values, adjustments and missing information for review.
10. The user confirms the submission.
11. The system saves the current values and displays the synchronisation and attachment-upload status.

Alternative Flows and Exceptions

A1 — Ticket information is incomplete. The user saves the known information. Missing fields remain explicitly identified. Unknown weights are not replaced with zero.

A2 — A reported weight is ambiguous. The system permits the source information to be retained as incomplete. The value is not used in calculations requiring an unconfirmed interpretation.

A3 — An adjustment is already applied. The system retains that indication and does not deduct the adjustment again.

A4 — Adjustment treatment is unknown. The information remains unresolved. The system does not assume that the adjustment must be applied or has already been applied.

A5 — Calculation rules are unconfirmed. The system stores reported values without generating an unsupported final weight or combined deduction.

A6 — The delivery contains a mixed load. Ticket totals remain associated with the delivery. They are not automatically divided among its properties or parcels.

A7 — A saved value requires correction. An authorised user reviews and confirms the replacement value. The system stores the current information without a permanent correction history and updates affected results.

A8 — A possible duplicate is detected. The system presents the available ticket and delivery information for review. A ticket number alone is not assumed to be globally unique.

A9 — The device is offline. Only non-financial structured ticket information is saved locally as pending synchronisation. Monetary fields and adjustments are excluded. Upload deferral does not authorise offline disclosure of monetary content in an attachment. Mixed-content file handling remains pending under Q-OFF-02.

A10 — Attachment upload fails. The system identifies the attachment as unavailable or pending. It does not imply that the source file has been stored on the server. The structured record retains its actual storage status.

Postconditions **On success:** Known ticket values and adjustments are saved against the selected delivery, with their meaning and units.

On incomplete entry: Unresolved information remains visible and is not treated as confirmed calculation input.

On correction: Current values replace the incorrect information, and affected totals are updated according to the approved calculation rules.

On offline save: Local changes and pending attachment uploads remain distinguishable from server acceptance.

On cancellation or rejected submission: Previously accepted information remains unchanged.

Related Requirements and Rules FR-049, FR-050, FR-054, FR-061

BR-012, BR-032, BR-035, BR-036.

Decision References Q-OFF-02, Q-DA-01, Q-DA-02, Q-DA-08, Q-AT-02.

4.20 UC-17 — Record Analysis Results

Purpose Allow an authorised user to record an olive-oil yield analysis associated with one delivery.

The system preserves the distinction between pending results, reported values and any application-calculated indicators.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant permissions.

The cooperative or laboratory supplies the analysis information. Direct integration is not assumed.

Preconditions

- The delivery exists.
- The user has permission for the requested operation.
- Offline entry requires the relevant authorised delivery information to be available on the device.

Complete analysis information is not required: a pending or incomplete entry may be saved.

Main Flow

1. The user selects a delivery and opens its analysis information.
2. The user creates an analysis entry or opens an existing one.
3. The user records the available analysis reference, reporting entity and analysis date.
4. The user enters the reported yield percentage where available.
5. The user records the reference weight and its meaning where known.
6. The user may attach an image or PDF of the source result.
7. The system validates the available information using the approved rules.
8. The system presents the reported values and identifies missing information.
9. The user reviews and confirms.
10. The system saves the current information and displays its synchronisation and attachment-upload status.

Alternative Flows and Exceptions

A1 — The result has not arrived. The entry remains pending. The system does not assign a zero yield percentage.

A2 — The result is known but reference information is incomplete. The system saves the reported result and identifies the missing information. Calculations requiring that information remain unavailable or explicitly incomplete.

A3 — An entry error is corrected. An authorised user reviews and confirms the corrected current values. The system updates affected indicators according to the approved rules, without maintaining detailed correction history.

A4 — A different analysis is received. The system distinguishes a new source analysis from a correction to an existing entry. Whether the new analysis replaces or supplements an earlier result follows the confirmed business policy. No automatic selection or double counting is assumed.

A5 — The campaign is closed. Analysis information for an existing delivery may be added or completed without reopening the campaign.

A6 — The device is offline. The system saves the available information locally as pending synchronisation. An attachment may remain pending upload until connectivity returns.

A7 — An attachment upload fails. The system identifies the attachment as pending or unavailable. Structured analysis information retains its actual storage status.

A8 — Information is invalid. The system identifies the affected values and preserves unaffected input. Validation uses approved value ranges and precision rules.

Postconditions **On success:** The analysis information is saved against the selected delivery, with its available source references.

On incomplete entry: Pending results and missing reference information remain explicitly identified.

On correction: Current values reflect the accepted correction. Dependent indicators use the updated information according to approved calculation rules.

On offline save: Local information and pending attachments remain distinguishable from server-accepted data.

On cancellation or rejection: Previously accepted information remains unchanged.

In all cases: Recording an analysis does not change the delivery date or overwrite official weighing values.

Related Requirements and Rules FR-051, FR-052, FR-054, FR-061

BR-012, BR-033, BR-034, BR-036, BR-038.

Decision References Q-DA-03, Q-DA-04, Q-DA-08, Q-OFF-02, Q-AT-02.

4.21 UC-18 — Manage Cooperative-Reported Ticket Values

Purpose Allow an authorised user to record, consult and correct the current monetary value reported by the cooperative for a weighing ticket.

The ticket identifies the associated delivery. The application does not calculate price estimates or process payments.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant monetary-data permissions.

Operational work permissions do not automatically grant access to monetary information.

The cooperative supplies the reported amount. Direct integration is not assumed.

Preconditions

- The weighing ticket exists and is associated with a delivery.
- The user has permission for the requested operation.
- Consultation, registration and correction of monetary information require an internet connection.
- For entry of an amount, source information is available.

A known amount is not required for consultation: the ticket may still be awaiting its value.

Main Flow

1. The user selects a delivery and its weighing ticket, or opens the ticket directly.
2. The system displays the current monetary information or identifies the value as awaiting receipt.
3. The user chooses to consult, record or correct the reported value.
4. For entry or correction, the user provides the amount in euros, reporting date and provisional or final status.
5. The system identifies the cooperative from the ticket context.
6. The user may add a source reference, observations and an image or PDF.
7. The system validates permissions, required information and ticket association.
8. The system presents the proposed current value for review.
9. The user confirms.
10. The system saves the current amount and status, replacing any previous value.
11. The system updates affected monetary indicators and confirms the result.

For consultation only, the flow ends after the available information is displayed.

Alternative Flows and Exceptions

A1 — No amount has been reported. The system identifies the ticket as awaiting its monetary value. It does not create a zero amount.

A2 — A final value replaces a provisional value. The system replaces the current amount and changes its status to final. The provisional amount is not retained as a separate historical value.

A3 — An entered value is incorrect. An authorised user reviews and confirms the corrected amount and status. Only the current information is retained.

A4 — Source details are incomplete. Optional references or observations may be added later. A reported amount must not be invented to complete an otherwise pending entry.

A5 — The ticket covers a mixed load. The amount remains associated with the ticket. It is not automatically allocated across properties or parcels.

A6 — The campaign is closed. The user may record or update the reported value of an existing ticket without reopening the campaign.

A7 — An attachment fails to upload. The system identifies the attachment failure separately from the amount's save status. It does not claim that the file was successfully stored.

A8 — Information is invalid or access is denied. The system rejects the affected operation and preserves recoverable input without exposing protected information.

A9 — Connectivity is unavailable. Monetary information cannot be consulted, registered or corrected offline. Prepared operational summaries and offline PDF reports exclude monetary information. Previously accepted monetary values remain unchanged.

A10 — Submission is repeated. Retrying the same operation does not create additional monetary contributions for the same ticket.

Postconditions **On successful entry or correction:** The ticket has one current reported amount, its provisional or final status and the required source information.

On replacement: The accepted amount replaces the previous value. Affected totals use the current value only.

On consultation: Current information is displayed, or the absence of a reported value is explicit.

On cancellation or rejected changes: Previously accepted information remains unchanged.

In all cases: The operation does not create a payment, alter ticket weights or modify analysis results.

Related Requirements and Rules FR-053, FR-054, FR-061

BR-012, BR-034, BR-035, BR-036, BR-040.

Decision References Q-DA-01, Q-DA-05, Q-DA-09, Q-OFF-02, Q-AT-02.

4.22 UC-19 — Consult Campaign Indicators

Purpose Allow an authorised user to understand the progress and results of a harvest campaign through operational, production and permitted monetary indicators.

The same consultation supports ongoing work and the final campaign review. Closing a campaign does not freeze its indicators.

The shared conditions in Section 4.2 apply; case-specific exceptions are defined below.

Actors **Primary actor:** Farm Owner.

Conditional actor: Field Manager, if included and granted the relevant consultation permissions.

Monetary indicators require separate monetary-data permission.

Preconditions

- The campaign exists.
- The user has permission to consult the requested scope.

- Offline consultation requires a previously prepared operational summary and relevant authorised local information.

Operational activity is not a precondition: a campaign may have no recorded work or deliveries yet.

Main Flow

1. The user selects one harvest campaign.
2. The system identifies the campaign and the authorised consultation scope.
3. The user selects the campaign overview, a property summary or parcel detail.
4. The user may apply date, worker and task filters where relevant.
5. The system retrieves or calculates the eligible indicators using approved rules.
6. The system presents totals, tables and useful simple charts.
7. Each indicator identifies its meaning, unit, scope and relevant completeness.
8. The user may open an indicator to inspect its contributing records.
9. The user may export the selected summary as a PDF.

Indicator Groups Work indicators:

- Distinct workers with confirmed daily presence and an explicitly identified actual assignment within the selected campaign.
- Distinct start dates with confirmed actual work periods.
- Total confirmed actual working hours.
- Hours by worker.
- Hours by task type: machinery operation and general work.
- Actual campaign assignments whose hours or task information remain incomplete.

Production indicators:

- Confirmed deliveries.
- Known weight totals, separated by weight meaning.
- Available yield results and approved yield aggregates.
- Pending analyses and relevant result coverage.

Monetary indicators:

- Current cooperative-reported ticket amounts.
- Separate provisional and final totals.
- Tickets awaiting monetary values.

Monetary consultation requires connectivity in the current version.

Alternative Flows and Exceptions

A1 — No matching activity exists. The system explains that no records match. Zero is used only where it is a valid result for the defined indicator. An undefined average is not shown as zero.

A2 — Data is incomplete. The system identifies pending information. Meaningful partial results show their known coverage and exclusions.

A3 — The campaign is closed. The user may consult its current results. Late analysis results, ticket information and reported values update the relevant indicators when accepted.

A4 — An accepted correction exists. Indicators use the current accepted values under the applicable correction rules. Previous values are not added as additional contributions.

A5 — Loads have mixed provenance. Known totals may contribute to the campaign summary. Property or parcel allocation is shown only when supported by known quantities or approved rules. Unallocated information remains identifiable.

A6 — Yield calculation is unsupported. The system displays available source results but does not invent reference weights or aggregation formulas.

A7 — Access covers only part of the campaign. Results are limited to the authorised scope. A restricted view is not labelled as the unrestricted campaign total.

A8 — The device is offline. The system displays the prepared operational summary and its last successful update. Monetary indicators are unavailable offline.

A9 — Local changes are pending. The operational summary combines the prepared server baseline with relevant local changes.

Creates are included once. Updates replace their corresponding baseline values. Locally accepted removals or cancellations remove the affected local contributions.

Affected indicators are identified as provisional until server acceptance. The interface does not imply knowledge of unsynchronised changes from other devices.

A10 — A pending change is rejected. The system distinguishes the unresolved proposal from server-accepted information.

The accepted summary is restored or recalculated from accepted values. Any still-visible local proposal is separately identified and is not presented as part of the accepted total.

A11 — Requested detail is unavailable offline. The system explains that connectivity is required for that detail. It does not present unavailable records as an empty result.

A12 — Retrieval or calculation fails. The system identifies the affected indicator and offers a retry where appropriate. Failure is distinguishable from zero or a successful query with no records.

A13 — A PDF is requested. The system exports the selected authorised summary with its campaign, filters, generation date and time, completeness and synchronisation limitations. The export represents that moment, rather than a permanently frozen campaign. An offline PDF contains operational information only. It excludes monetary indicators and identifies pending local contributions and the last successful synchronisation.

Postconditions **On success:** The user can inspect the current indicators for the selected authorised scope and access available supporting records.

On offline consultation: The displayed operational summary identifies its synchronisation date and pending local changes.

On export: A PDF reflects the selected summary and its relevant limitations at generation time.

In all cases: Consultation does not modify source records, create payments or close the campaign.

Related Requirements and Rules FR-055, FR-056, FR-057, FR-058, FR-059, FR-060, FR-061, FR-062

BR-014, BR-018, BR-019, BR-020, BR-035, BR-036, BR-037, BR-038, BR-039, BR-040.

Decision References Q-DA-01, Q-DA-05, Q-IND-01, Q-IN-02.

4.23 UC-20 — Generate Worker Documents

Status Scope awaiting confirmation with the team and client.

The use case identifier is reserved. The detailed generation workflow is not yet an approved requirement.

Candidate Purpose Generate selected worker documents from agreed templates and reviewed information, reducing repeated manual entry.

This capability is distinct from uploading existing files through UC-13 and signing a version through UC-14.

Candidate Actors Farm Owner.

Field Manager only if the role is included and the relevant generation permissions are granted.

Consistency Constraints If generation is included:

- The user must review the document before accepting it.
- Generation must not automatically mark the document as signed.
- Editing generation inputs must not silently modify source worker records.
- The accepted output must identify its document category and version.
- Access and version retention must align with UC-12 and UC-13.
- Signing must follow UC-14 for the specific reviewed version.

Acceptance Status Detailed acceptance criteria and implementation estimates remain pending scope confirmation.

Related Requirements and Rules Generation requirements remain pending under UC-20.

Decision References Q-DOC-01.

See Q-DOC-01 for templates, data fields, output, offline scope and version behaviour. No generation requirement is approved by this placeholder.

5 Functional Requirements

The catalogue groups related behaviours into FR-001 to FR-062. Lettered acceptance criteria preserve independently verifiable conditions. Catalogue inclusion does not establish MVP priority or stakeholder approval. Manager-specific behaviour is conditional on Q-AA-01; access revalidation also applies to the owner. Document generation remains pending under UC-20.

5.1 Requirements Catalogue

Common acceptance conditions. The shared conditions in Section 4.2 apply to every relevant requirement: business/operation authorisation, accurate local/server status, retry without duplicate effects, preservation of accepted data on cancellation, recoverable validation failures and explicit handling of unknown values. Connectivity remains operation-specific. The server revalidates offline proposals, including access and conflicts.

Criterion references for verification. Verification checks and implementation tasks identify the requirement and the relevant lettered acceptance criterion. For example, FR-041(a) covers acceptance of a new current version, FR-041(b) covers retention and deletion, and FR-041(c) covers incorrect uploads and withdrawal.

5.1.1 Registration, Authentication and Access

ID	Functional requirement	Acceptance criteria
FR-001	The system shall register an owner and establish a business workspace.	(a) Request name, email, password and business name; validate the input and prevent duplicate email identities using one consistent normalisation policy. (b) Verify the email before granting active business access; support replacement verification messages and reject invalid or expired links. (c) Activation creates one owner workspace without operational records. Retries do not create duplicate accounts or workspaces.
FR-002	The system shall authenticate account holders and enforce business access.	(a) Activated accounts can log in with email and password. Invalid credentials do not unnecessarily disclose account information; unverified accounts cannot access the workspace. (b) Validate session, business membership, responsibility scope and operation-specific permissions for protected requests, including direct record and file access. An expired session requires renewed authentication.

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ID	Functional requirement	Acceptance criteria
FR-003	The system shall terminate sessions and isolate local data between accounts.	(a) Logout is available online and offline and blocks local workspace access. (b) Identify pending work, offer synchronisation when online and preserve recoverable pending changes in protected storage. Another account cannot access or submit those changes. (c) The originating account requires renewed authentication and authorisation before normal recovery or submission; server-side logout actions requiring connectivity are completed on reconnection.
FR-004	The system shall support authorised offline continuity and access revalidation.	(a) Initial login and renewal of offline authorisation require internet. Offline continuation uses previously authenticated, prepared data and only explicitly supported operations. (b) A bounded offline authorisation applies independently of the two-week preparation window. Reconnection revalidates current access before normal acceptance of pending work. (c) Revoked-manager proposals follow FR-008; offline expiry and security parameters remain under Q-AA-04 and Q-AA-05.
FR-005	The system shall allow users to recover their account access through a verified reset process.	(a) A recovery request gives a generic acknowledgement; eligible accounts receive a reset message. (b) Only a valid, unexpired, unused authorisation permits a password satisfying the configured policy. A successful reset consumes it and invalidates existing online sessions. (c) Ownership and permissions remain unchanged. Disconnected sessions detect invalidation on reconnection or expiry of offline authorisation. Recovery without inbox access is not assumed.
FR-006	The system shall invite managers and manage invitation acceptance.	(a) The owner specifies recipient email, business scope and permissions and may inspect, resend or cancel pending invitations. (b) Acceptance requires the intended verified account and a valid invitation. New invitees may create an account without automatically creating an owner workspace. (c) Existing memberships and repeated acceptance or resend requests do not create duplicate memberships; expired or cancelled invitations do not grant access.

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ID	Functional requirement	Acceptance criteria
FR-007	The system shall configure and enforce delegated scope and permissions.	(a) Only the owner administers access. A manager cannot extend their own authority or grant access to others. (b) Distinguish consultation, planning, entry, corrections, document operations, signing initiation and financial permissions. Insurance consultation is required for workers within the assigned responsibility. (c) Saved changes govern subsequent online requests. Role inclusion and scope granularity remain conditional under Q-AA-01 and Q-AA-02.
FR-008	The system shall revoke manager access and review existing pending proposals.	(a) Confirmed revocation blocks subsequent normal online business access without deleting the account or business records. Disconnected devices enforce it on reconnection or offline expiry. (b) Existing pending proposals may be recovered for owner review without restoring manager access. The owner may accept or reject them after date, lifecycle, permission, coverage and conflict validation. (c) Retain the originating account and accepting owner separately. New unauthorised operations must not masquerade as previously pending work; unresolved proposals are not silently discarded.

5.1.2 Properties and Campaigns

ID	Functional requirement	Acceptance criteria
FR-009	The system shall create properties with required and optional descriptive information.	(a) Name and textual location are required; declared hectares and one or more olive varieties are optional and can be supplied later. (b) Unknown area remains unknown. Ownership is associated with the workspace rather than repeatedly entered as free text. A map is not required.
FR-010	The system shall create named parcels within a property.	(a) Require a name and an authorised parent property; area and varieties are optional. (b) A property may have no parcels. Parcel creation during campaign setup uses the same permission and creates a reusable property parcel, not a campaign-only division.
FR-011	The system shall allow authorised users to consult and update properties and parcels.	(a) Updates preserve identity and historical associations; new parcels do not redistribute historical property-level records. (b) Prepared authorised details are consultable offline. Creation and changes require connectivity; missing local data is not evidence that a record does not exist.

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ID	Functional requirement	Acceptance criteria
FR-012	The system shall archive and reactivate properties and parcels.	(a) Confirmed archival preserves history and prevents new operational associations, including through parcels of an archived property. (b) Reactivation requires connectivity and authority. Parcel availability depends on its parent; reactivating a parent does not reactivate independently archived parcels.
FR-013	The system shall delete eligible properties and parcels.	(a) Deletion requires permission, connectivity, explicit confirmation and no associated records. (b) Otherwise refuse deletion and offer archival. Cancelling preserves the record.
FR-014	The system shall create campaigns and select their area coverage.	(a) Require a name, planned start and at least one authorised whole property or parcel; several properties are permitted. (b) Prevent duplicate parent-property and parcel coverage. Parcels can be created through FR-010 before returning to selection.
FR-015	The system shall allow authorised users to consult and update campaigns and coverage.	(a) Prepared campaign details are consultable offline; management changes require connectivity. (b) Validate added coverage. Before removing an area with dependent records, reopen a closed campaign and resolve dependencies through permitted correction or deletion workflows. No cascading deletion occurs; unresolved dependencies block removal.
FR-016	The system shall maintain distinct planned and actual campaign dates.	(a) An authorised user may enter actual start manually. If none exists, the first confirmed actual campaign-area assignment proposes its working date for explicit confirmation or correction. (b) Planning or daily presence alone does not establish actual start. Offline work may remain pending while the date awaits connected confirmation; existing actual dates are not silently changed. (c) Record actual end on closure and validate dates against activity. Editing planned start does not overwrite actual dates.
FR-017	The system shall close and reopen campaigns.	(a) Closure requires an actual end date, validation and connected authorisation; preserve existing records. (b) Reopening requires its own permission, date review and overlap validation. Completion, correction and late external information follow BR-011 and BR-012.

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ID	Functional requirement	Acceptance criteria
FR-018	The system shall archive, unarchive or delete eligible campaigns.	(a) Archival changes usual-list visibility independently of open or closed state and does not release coverage. Unarchiving does not reopen. (b) Delete only without operational records, with permission, connectivity and confirmation; otherwise offer archival.
FR-019	The system shall prevent conflicting campaign coverage.	(a) An open campaign blocks another campaign covering the same area, including one planned to start in the future. (b) A whole property conflicts with its parcels; campaigns on different parcels may coexist. Archival does not release coverage and historical dates remain relevant for closed campaigns. (c) Coverage edits, parcel changes and reopening cannot bypass overlap checks.

5.1.3 Workers, Planning and Actual Work

ID	Functional requirement	Acceptance criteria
FR-020	The system shall create and update business-specific worker records.	(a) Require full name and telephone; assign an internal identifier without creating an application account. (b) Record machinery capability and an optional description separately from actual tasks. Edits preserve associations and historical task attribution; retain current profile values and last-modified date. (c) Warn on matching names or telephones within the business; permit confirmed distinct records without automatic merging or exposure of other businesses. Management requires connectivity.
FR-021	The system shall inactivate and reactivate workers.	(a) Preserve work and documents; remove inactive workers from usual new-work selection while flagging existing future plans without silently changing them. (b) Reactivation reuses the identity. Not being selected or attending on a date does not inactivate a worker.
FR-022	The system shall delete eligible worker records.	(a) Require no associated records, connected authorisation and explicit confirmation. (b) Refuse deletion when dependencies exist and offer inactivation; cancelling preserves the worker.

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ID	Functional requirement	Acceptance criteria
FR-023	The system shall allow authorised users to consult authorised worker records online and offline.	(a) Prepared current profiles are consultable offline, with availability limitations identified. (b) Access to a profile does not grant access to every associated document. Creation and profile management remain online.
FR-024	The system shall prepare independent daily work plans with group and individual values.	(a) Select campaign, eligible area, workers and one or more target dates; create independent plans for each date. (b) Hours and tasks may be omitted for one area per worker and working date. Several areas require periods for all relevant assignments. (c) Apply periods, tasks and optional counted or non-counted breaks to selected workers, allowing individual adjustments without altering others. Plans do not confirm attendance or work.
FR-025	The system shall reuse a plan or actual work as a reviewed planning proposal.	(a) Propose workers, areas, periods, tasks and applicable breaks for selected target dates without modifying source records or copying attendance and actual confirmation. (b) Validate each target independently; existing plans are not silently overwritten or duplicated. Inactive workers require review and eligible replacement or reactivation.
FR-026	The system shall validate planned periods and scheduling conflicts.	(a) Defined periods must be valid and non-overlapping for the worker across properties and campaigns, including adjacent-date overnight intervals. (b) Full start and end timestamps determine duration and start-date association. Missing hours do not establish a successful conflict-free validation. (c) Planned and actual interval validation is separate; matching a plan and its actual outcome does not itself duplicate actual work.
FR-027	The system shall adjust eligible plans while preserving their role as an expectation.	(a) Workers may be removed or replaced before actual work exists in the affected daily area context. (b) Once actual work exists there, preserve the corresponding plan as a reference; differences belong to actual entry or correction. Multi-day target plans remain independently editable.

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ID	Functional requirement	Acceptance criteria
FR-028	The system shall allow authorised users to consult schedules through daily, worker and weekly views, including offline preparation.	(a) Views use the same plans and support campaign, property, parcel, worker and task filters; identify undefined hours/tasks, breaks, inactive workers and overnight end dates. (b) Prepare authorised schedules for the current and following week; additional dates require explicit preparation. Display last synchronisation and pending changes. (c) Distinguish an empty successful query from retrieval failure or unavailable local data. Consultation does not confirm work or attendance.
FR-029	The system shall create and adjust plans offline using prepared reference data.	(a) Save supported changes locally with pending status and revalidate permissions and business rules on reconnection. (b) Rejected or conflicting changes are not labelled accepted; recovery preserves source and target-plan integrity under the common conditions.
FR-030	The system shall confirm daily attendance and actual campaign-area assignments.	(a) Use one attendance decision per worker and start-based working date in the business: awaiting confirmation, present or absent. No decision does not mean absence. (b) Allow existing unplanned workers and attendance without complete hours/tasks. Actual participation requires an explicitly confirmed campaign-area assignment. (c) Absence cannot coexist with contradictory actual assignments or periods; correcting a confirmed decision follows UC-11. Driver selection alone does not establish attendance.
FR-031	The system shall record and explicitly confirm actual periods, tasks and work areas.	(a) Review planned proposals or enter actual work directly, including without a plan; group confirmation permits individual adjustments and leaves the plan unchanged. (b) Record separate periods for changes of actual area or task. A valid interval with an unknown task contributes as unclassified rather than being omitted. (c) Use complete timestamps and the start-date association for overnight periods. Actual assignment, interval completeness and attendance remain distinct.

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ID	Functional requirement	Acceptance criteria
FR-032	The system shall record breaks and calculate counted working time.	(a) A break identifies its worker, start, end and counting treatment. Only non-counted time intersecting that same worker's actual periods reduces hours. (b) Counted breaks and gaps outside recorded periods are not deducted. Overlapping non-counted break time is deducted once, with the applicable share attributed to each task period. (c) Use the formula and examples in Section efsec:counted-working-time; no payroll calculation is implied.
FR-033	The system shall validate actual periods and prevent worker-time overlap.	(a) Reject invalid or overlapping actual intervals for the same worker across areas, campaigns and adjacent dates. (b) Consider the full overnight interval. Planned periods are not treated as additional actual work.
FR-034	The system shall record attendance and actual work offline.	(a) Prepared workers and campaign context support locally confirmed attendance, assignments, periods and breaks. (b) Expose local storage and server acceptance separately. Retry and reconnection follow common validation and duplicate-prevention conditions.
FR-035	The system shall enter retrospective work and complete missing information.	(a) Preserve actual working date separately from entry timestamp. (b) A closed campaign permits previously missing details or periods within valid actual dates if confirmed values and associations remain unchanged and attendance, coverage and overlap checks pass. (c) Changing or removing confirmed work requires reopening. Work outside valid actual dates requires reopening and date review; entry does not silently change campaign dates.
FR-036	The system shall correct attendance, actual work and valid record associations.	(a) Require specific correction permission for all affected origin and destination contexts; validate dates, coverage, overlaps and open status of affected campaigns. No age-based deadline applies. (b) Permit correction of worker, date, campaign, area, periods, task and breaks without rewriting the plan or unrelated work. (c) Changing presence to absence requires confirmation of removal of all contradictory actual assignments, including those without hours, periods and breaks across the business working date. Cancelling preserves them. (d) Moving one period must preserve other work and daily-attendance consistency. Current values are retained without permanent field-change history.

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ID	Functional requirement	Acceptance criteria
FR-037	The system shall support offline corrections and authorised conflict resolution.	(a) Save prepared corrections as pending. If accepted server values changed, present them beside the proposal for authorised manual review. (b) The user may keep server values or submit a reviewed correction; validate again before acceptance. Temporary comparison is not permanent correction history.

5.1.4 Worker Documents and Signing

ID	Functional requirement	Acceptance criteria
FR-038	The system shall allow authorised users to consult and download authorised worker documents and retained versions.	(a) List contracts, insurance and other documents with title, version and worker-specific signature information. Open and download supported available PDF, JPG and PNG files. (b) The owner's access covers the business; managers require the document and worker scope defined by BR-024. Profile access alone is insufficient. (c) Expose retained versions under BR-026 and expiry under BR-025, including unknown validity, expired and withdrawn versions. Current does not mean valid or signed; a failed or unavailable retrieval is not an empty list.
FR-039	The system shall prepare insurance documents for field consultation.	(a) Prepare retained current and previous versions for workers planned in the current and following week, plus additional existing workers explicitly selected while connected. (b) Identify preparation failure, missing local files and last synchronisation. Apply received deletion and access changes on reconnection; the window does not define offline-session duration.
FR-040	The system shall upload worker documents and their metadata.	(a) Require online modification permission, one or more existing same-business workers, category, title and file; observations are optional. Record upload date and version identity automatically. (b) Accept PDF, JPG or PNG within the proposed 10 MB limit, subject to Q-DOC-06. Unknown insurance expiry remains unconfirmed. (c) One shared file may serve several workers. External-signature declaration is separate from application evidence and does not assign signatures to all workers. (d) Validate and review the upload before acceptance; attachment content is not extracted automatically.

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ID	Functional requirement	Acceptance criteria
FR-041	The system shall manage document versions, retention, withdrawal and permitted deletion.	(a) A successfully accepted new version becomes current without inheriting signatures. Cancellation, failure and duplicate retries do not replace versions or trigger extra cleanup. (b) Apply category retention in BR-026. Delete older insurance/other versions even if signed, covering source files, derivatives and version-specific signature data across shared associations and synchronised local copies. (c) Delete an incorrect unsigned version only without dependencies and after authorisation and confirmation. Withdraw signed incorrect versions while retained; a withdrawn current version stays current and withdrawn until replaced, without automatic fallback. (d) File-content replacement requires a new version, not metadata editing. Management is online; disconnected copies receive deletion on synchronisation, not immediate remote wipe.
FR-042	The system shall conduct in-person document review and explicit worker signing.	(a) An authorised initiator identifies the physically present worker and eligible document version; no worker account or owner signature is required. (b) The worker reviews the document, draws with finger or pen, may clear/redraw or cancel before confirmation, and explicitly confirms a non-empty drawing and reviewed version. (c) Record worker, version, initiator and date/time. The confirmed drawing is not replaced or transferred to another version; upload or generation does not create a signing event.
FR-043	The system shall prevent duplicate signing and recover uncertain submissions.	(a) One worker does not repeat completed signing of the same version. Display accepted results and check/resume pending or uncertain operations before asking for another signature. (b) Distinguish no confirmed signature, local pending, accepted and resolution-required outcomes. Preserve recoverable results and retry without duplicate events.
FR-044	The system shall maintain individual signing results for shared documents.	(a) Track completion by worker and version; one signature does not sign for others. (b) The worker-specific application view shows only that worker's captured signature. This does not redact personal information or external signatures already embedded in the source file; source access follows approved sharing permissions.

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ID	Functional requirement	Acceptance criteria
FR-045	The system shall prepare and synchronise offline signing for the reviewed version.	(a) Explicitly prepared authorised documents, including contracts, may be reviewed and signed offline. Do not automatically download all contracts. (b) Keep locally confirmed signatures pending. Accept against the original version only if it remains eligible, showing a newer version where applicable. (c) A withdrawn or deleted original requires resolution; do not transfer the signature or recreate a retention-deleted version. Detailed resolution remains under Q-DOC-04.

5.1.5 Deliveries, Weighing, Analyses and Reported Values

ID	Functional requirement	Acceptance criteria
FR-046	The system shall register and ordinarily correct transport and load provenance.	(a) Require campaign, at least one eligible property/parcel, registered worker driver, vehicle, destination and departure timestamp. Selection as driver does not create attendance or hours. (b) Permit several source areas without duplicate property/parcel quantities; collection origin and varieties may be unknown. Reuse a destination or enter a new name without a separate management module. (c) No weight, analysis or amount is required. Ordinary correction validates timestamps and associations within the same campaign and state; cancellation, reversal and cross-campaign reassignment remain under Q-DA-06.
FR-047	The system shall confirm a delivery or retrospectively register the complete transport.	(a) Save departure as in transport, then let an authorised user confirm delivered with actual arrival timestamp not before departure. (b) A completed journey may be entered directly as delivered with both timestamps. Event timestamps remain distinct from entry timestamps; one associated transport per delivery is the current proposal.
FR-048	The system shall record departures and delivery confirmations offline.	(a) Use prepared campaign, area and worker data, allowing a new destination name. (b) Physical state remains separate from pending synchronisation. Validate current authority and campaign context before server acceptance.

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ID	Functional requirement	Acceptance criteria
FR-049	The system shall record weighing tickets, reported adjustments and source attachments.	(a) Each ticket belongs to one delivery; multiplicity remains under Q-DA-01. Record available source reference, cooperative, weighing timestamp and weight meanings/units without assuming global reference uniqueness. (b) Incomplete or ambiguous source values remain explicit and are excluded from calculations requiring a confirmed meaning. Record adjustments and whether already applied; do not invent final weights or reapply deductions. (c) Allow PDF/image evidence without automatic extraction. Structured data may be entered offline; upload status remains separate and recoverable. Monetary adjustments and mixed-content files follow the financial access condition and Q-OFF-02.
FR-050	The system shall correct current weighing and adjustment information.	(a) An authorised correction replaces current values without permanent field-change history. A genuinely additional or replacement ticket is governed by Q-DA-01, not silently added to totals. (b) Apply updated eligible values to indicators through FR-061. Concurrent changes follow the common conflict principle.
FR-051	The system shall record analysis results, incomplete information and supporting files.	(a) Associate each analysis with one delivery; multiplicity and selection remain under Q-DA-03. Record available yield percentage, reporting entity, reference, analysis date and reference weight with its meaning; distinguish entry date. (b) A pending yield is not zero and a missing reference weight is not inferred. Permit PDF/image evidence without extraction. (c) Allow prepared offline structured entry with separate pending-attachment status. Eligible aggregation requires approved selection and reference-weight rules.
FR-052	The system shall correct an analysis while distinguishing a new source result.	(a) Authorised entry corrections replace current values without permanent field-change history and refresh eligible indicators. (b) A genuinely new analysis is not silently merged into a correction; replacement or supplementary-result policy remains under Q-DA-03. Analysis entry does not change delivery dates or official weights.

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ID	Functional requirement	Acceptance criteria
FR-053	The system shall allow authorised users to consult and manage the current cooperative-reported ticket amount.	(a) Require online financial permission and an existing ticket. Store EUR with two decimal places, reporting cooperative/date and provisional or final classification; reference, observations and PDF/image evidence are optional. (b) A final amount replaces the provisional amount; corrections retain one current value, without permanent prior-value history. Missing is not zero, and the same record is reached through ticket or delivery. (c) Do not calculate prices, reapply deductions, infer payment or allocate mixed-load amounts without approved information. Attachment failure is distinct from amount-save failure; source attachments do not inherit worker-document retention. (d) Exclude monetary data from offline datasets and reports; mixed-content attachment treatment remains under Q-OFF-02. Final-to-provisional reclassification remains under Q-DA-09.
FR-054	The system shall complete and correct external information after campaign closure.	(a) Authorised users may add or correct ticket information, analyses and reported amounts for existing deliveries/tickets without reopening. (b) This exception does not permit correction of confirmed work, new operational activity outside valid dates, delivery reversal or campaign reassignment.

5.1.6 Campaign Indicators and Reports

ID	Functional requirement	Acceptance criteria
FR-055	The system shall allow authorised users to consult campaign indicators and property/parcel breakdowns.	(a) Consult one authorised campaign during or after activity, with campaign overview, primary property summaries and parcel detail where available. (b) A campaign need not already contain activity. Preserve record identity and distinguish partial authorised scope from an unrestricted campaign total; cross-campaign comparisons remain outside baseline.
FR-056	The system shall filter indicators and inspect their supporting records.	(a) Support dates, property, parcel, worker and task where a defined source relationship exists. Label each indicator's applicable scope and date basis. (b) Open contributing records only within permission and local availability. A worker/task filter does not implicitly allocate deliveries or money; offline-unavailable detail is not an empty result.

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ID	Functional requirement	Acceptance criteria
FR-057	The system shall present actual participants, working days and counted hours.	(a) Count distinct workers with confirmed presence and actual campaign assignment, even if hours are incomplete. Count distinct start dates with eligible actual periods. (b) Show counted hours overall, by worker and task, including unclassified hours. Use actual confirmation and worker-specific breaks, not plans or machinery capability. (c) Distinct workers/dates are recalculated for campaign totals, not summed across areas; unknown intervals and incomplete assignments remain visible.
FR-058	The system shall present eligible delivery, weight and analysis indicators.	(a) Count distinct confirmed deliveries, not tickets. Separate weights by meaning/unit and prevent duplicate or unsupported source aggregation. (b) Show reported yield and an aggregate only under approved eligibility/reference-weight rules. Missing results are not zero; undefined aggregates remain unavailable and partial results disclose coverage. (c) Mixed loads contribute once to eligible campaign totals; unallocated weights/amounts remain identifiable rather than being copied to every source area. Theoretical oil is not actual oil received.
FR-059	The system shall present current reported monetary indicators.	(a) With online financial permission, show separate eligible provisional and final totals and tickets awaiting an amount. (b) Use each eligible ticket's current amount once; replacements are not additional contributions. Aggregation of several tickets for one delivery requires confirmed source semantics; money does not establish payment.
FR-060	The system shall present complete or partial operational indicators offline.	(a) Expose missing/unallocated data, unresolved rules, available local scope, last synchronisation and pending contributions. Do not claim knowledge of later changes on other devices. (b) Combine the prepared accepted baseline with local proposals: creates contribute once, updates replace and removals suppress their prior contribution. Affected results are provisional. (c) Missing local data, retrieval failure, undefined results and valid zero are distinguishable. Monetary indicators are unavailable offline.

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ID	Functional requirement	Acceptance criteria
FR-061	The system shall refresh derived totals after source changes without duplicate contributions.	(a) Recalculate affected origin and destination totals after accepted source corrections or late information, including after closure. Indicators cannot be edited directly. (b) Supported local proposals update provisional results, not accepted central totals. Rejected proposals do not remain represented as accepted; retain any unresolved proposal separately. (c) Campaign closure does not freeze results. Previous exported reports remain snapshots.
FR-062	The system shall export an authorised campaign indicator snapshot as PDF.	(a) Include campaign, authorised scope, applicable filters/date basis, generation timestamp and relevant completeness or synchronisation limitations. (b) Offline output includes operational information only, identifying last synchronisation and pending contributions; online financial content requires financial permission. (c) Export does not alter source records. Detailed Excel export and dedicated printing remain future options.

5.2 Mapping to Use Cases

The mapping identifies the use cases responsible for each functional requirement. Other use cases may consume the resulting records without redefining the capability.

5.2.1 Registration, Authentication and Access

ID	Use cases	Capability
FR-001	UC-01	Register an owner and establish a business workspace.
FR-002	UC-02	Authenticate account holders and enforce business access.
FR-003	UC-02	Terminate sessions and isolate local data between accounts.
FR-004	UC-02, UC-04	Support authorised offline continuity and access revalidation.
FR-005	UC-03	Recover account access through a verified reset process.
FR-006	UC-04	Invite managers and manage invitation acceptance.
FR-007	UC-04	Configure and enforce delegated scope and permissions.
FR-008	UC-04	Revoke manager access and review existing pending proposals.

5.2.2 Properties and Campaigns

ID	Use cases	Capability
FR-009	UC-05	Create properties with required and optional descriptive information.
FR-010	UC-05, UC-06	Create named parcels within a property.
FR-011	UC-05	Consult and update properties and parcels.
FR-012	UC-05	Archive and reactivate properties and parcels.
FR-013	UC-05	Delete eligible properties and parcels.
FR-014	UC-06	Create campaigns and select their area coverage.
FR-015	UC-06	Consult and update campaigns and coverage.
FR-016	UC-06, UC-10	Maintain distinct planned and actual campaign dates.
FR-017	UC-06	Close and reopen campaigns.
FR-018	UC-06	Archive, unarchive or delete eligible campaigns.
FR-019	UC-06	Prevent conflicting campaign coverage.

5.2.3 Workers, Planning and Actual Work

ID	Use cases	Capability
FR-020	UC-07	Create and update business-specific worker records.
FR-021	UC-07	Inactivate and reactivate workers.
FR-022	UC-07	Delete eligible worker records.
FR-023	UC-07	Consult authorised worker records online and offline.
FR-024	UC-08	Prepare independent daily work plans with group and individual values.
FR-025	UC-08	Reuse a plan or actual work as a reviewed planning proposal.
FR-026	UC-08	Validate planned periods and scheduling conflicts.
FR-027	UC-08	Adjust eligible plans while preserving their role as an expectation.
FR-028	UC-09	Consult schedules through daily, worker and weekly views, including offline preparation.
FR-029	UC-08	Create and adjust plans offline using prepared reference data.

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ID	Use cases	Capability
FR-030	UC-10	Confirm daily attendance and actual campaign-area assignments.
FR-031	UC-10	Record and explicitly confirm actual periods, tasks and work areas.
FR-032	UC-10	Record breaks and calculate counted working time.
FR-033	UC-10, UC-11	Validate actual periods and prevent worker-time overlap.
FR-034	UC-10	Record attendance and actual work offline.
FR-035	UC-10	Enter retrospective work and complete missing information.
FR-036	UC-11	Correct attendance, actual work and valid record associations.
FR-037	UC-11	Support offline corrections and authorised conflict resolution.

5.2.4 Worker Documents and Signing

ID	Use cases	Capability
FR-038	UC-12	Consult and download authorised worker documents and retained versions.
FR-039	UC-12	Prepare insurance documents for field consultation.
FR-040	UC-13	Upload worker documents and their metadata.
FR-041	UC-12, UC-13, UC-14	Manage document versions, retention, withdrawal and permitted deletion.
FR-042	UC-14	Conduct in-person document review and explicit worker signing.
FR-043	UC-14	Prevent duplicate signing and recover uncertain submissions.
FR-044	UC-12, UC-14	Maintain individual signing results for shared documents.
FR-045	UC-14	Prepare and synchronise offline signing for the reviewed version.

5.2.5 Deliveries, Weighing, Analyses and Reported Values

ID	Use cases	Capability
FR-046	UC-15	Register and ordinarily correct transport and load provenance.
FR-047	UC-15	Confirm a delivery or retrospectively register the complete transport.
FR-048	UC-15	Record departures and delivery confirmations offline.
FR-049	UC-16	Record weighing tickets, reported adjustments and source attachments.
FR-050	UC-16	Correct current weighing and adjustment information.
FR-051	UC-17	Record analysis results, incomplete information and supporting files.
FR-052	UC-17	Correct an analysis while distinguishing a new source result.
FR-053	UC-18	Consult and manage the current cooperative-reported ticket amount.
FR-054	UC-16, UC-17, UC-18	Complete and correct external information after campaign closure.

5.2.6 Campaign Indicators and Reports

ID	Use cases	Capability
FR-055	UC-19	Consult campaign indicators and property/parcel breakdowns.
FR-056	UC-19	Filter indicators and inspect their supporting records.
FR-057	UC-19	Present actual participants, working days and counted hours.
FR-058	UC-19	Present eligible delivery, weight and analysis indicators.
FR-059	UC-19	Present current reported monetary indicators.
FR-060	UC-19	Present complete or partial operational indicators offline.
FR-061	UC-11, UC-16, UC-17, UC-18, UC-19	Refresh derived totals after source changes without duplicate contributions.
FR-062	UC-19	Export an authorised campaign indicator snapshot as PDF.

6 Business Rules and Record States

The forty consolidated rules define shared constraints once. States, correction principles and formulas below describe their application. Pending questions and reviewed decisions are centralised in Section 10.

6.1 Business Rules

ID	Business rule
BR-001	Self-registration by an owner requires name, email, password and business name. One consistently normalised email identity identifies one account. Verification precedes active access and creates one owner workspace containing several properties and campaigns. Account, business membership and worker record are distinct; workers have no management accounts and invited managers do not automatically receive an owner workspace.
BR-002	Authentication does not replace business and operation authorisation. Initial login, activation, recovery and access administration require connectivity. Offline access requires previous authentication and bounded authorisation. Revocation or reset is enforced online and on disconnected devices at reconnection or offline expiry. Logout protects retained pending data from other accounts. Reset authorisations are single-use and time-limited; reset changes neither ownership nor permissions and its responses do not disclose account existence.
BR-003	Only the owner administers manager invitations and permissions. The intended verified account explicitly accepts a valid invitation; pending, cancelled or expired invitations confer no access. Consultation, entry, correction, document, signing and financial permissions are distinct. Managers responsible for workers must consult their insurance within that scope; role and granularity remain conditional.
BR-004	Pending proposals are revalidated on reconnection. Revoked-manager proposals require owner acceptance or rejection through a recovery workflow that does not restore business access. Normal business checks still apply. Preserve the originating account and accepting owner; do not accept new unauthorised work disguised as pending work or silently discard unresolved proposals.
BR-005	A property may have no parcels; each parcel belongs to one property. Declared hectares and varieties are optional. Declared area is the reference when known; map estimates are separate future information. Property-level quantities are not redistributed among parcels and parcel-area totals are not assumed to equal the whole property.
BR-006	Property/parcel deletion requires no associated records; otherwise use archival. Campaign deletion requires no operational records. Archival preserves history. An archived property blocks new operational use through itself and its parcels; reactivating the parent does not reactivate independently archived parcels.
BR-007	A campaign covers at least one whole property or eligible parcel and may cover several properties. Parent and child areas must not duplicate coverage. Parcels created during setup remain reusable property records. New divisions do not silently redistribute historical records.

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ID	Business rule
BR-008	An open campaign blocks other campaigns on the same area, even with future planned starts. Whole-property coverage conflicts with its parcels; different parcels may coexist. Campaign archival affects visibility independently of open/closed state and never releases coverage. Closed historical dates and reopening remain subject to overlap checks.
BR-009	Planned start is required. Actual start is separate, entered manually or proposed from the first confirmed actual campaign-area assignment when absent; confirmation requires connectivity. Presence or planning alone does not start a campaign. Existing actual dates are not silently changed; actual end is recorded at closure.
BR-010	Remove campaign coverage only after resolving dependent records through permitted correction or deletion workflows. Reopen a closed campaign before changing coverage or operational associations. Validate affected contexts and do not cascade-delete records; unresolved or undefined corrections block removal.
BR-011	Closed campaigns permit missing work details or previously unrecorded periods within valid actual dates, without replacing/removing confirmed values or changing confirmed associations. Attendance, coverage and overlap must remain consistent. Correcting/removing confirmed work requires reopening, with no age-based deadline. Work outside valid actual dates requires reopening and date review.
BR-012	Ticket information, analyses and reported amounts for existing deliveries/tickets may be added or corrected after closure without reopening. This exception does not authorise new operational activity outside campaign dates or exceptional delivery changes.
BR-013	Property, parcel, campaign, worker-profile and document-version management require connectivity. Prepared consultation and specifically defined field operations may work offline. All operations require relevant business scope and permission; offline availability does not imply unrestricted application access.
BR-014	Worker identity is business-specific and reused across its properties/campaigns, without cross-business merging. Matching name or telephone gives a warning, not mandatory rejection. Profiles retain current values and last-modified date rather than permanent field-change history; identity and operational associations survive edits.
BR-015	Machinery-operation capability is profile information, not the task performed. An operator may do general work. Each actual period records its known task or remains unclassified; profile capability changes do not relabel historical periods.
BR-016	Worker inactivation preserves history and flags existing future plans for review rather than deleting them. New assignments require an eligible worker or authorised reactivation. Non-selection or absence on a date is not inactivation.
BR-017	Multi-date plans are independent. Copying a plan or actual work produces a reviewed proposal, never attendance or actual confirmation. Hours/tasks are optional for one area per worker/date; multiple areas require explicit periods. Actual work freezes the corresponding daily-area plan as an expectation. Defined planned intervals must not overlap; validate planned and actual intervals separately.

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ID	Business rule
BR-018	Attendance is one decision per worker/start-based working date in the business: awaiting confirmation, present or absent. Campaign-area participation is separate and may have incomplete hours/tasks. No confirmation does not mean absence. Planned periods and breaks require explicit actual confirmation; absence cannot coexist with contradictory actual assignments.
BR-019	Actual area/task changes require distinguishable periods and a worker's actual intervals cannot overlap across contexts. Use full timestamps for positive duration and adjacent-date checks. Overnight periods belong to their start date, distinct from entry date.
BR-020	Only explicitly non-counted breaks belonging to the same worker reduce intersecting actual periods. Use the union of such intervals to deduct time once. Counted breaks and gaps already outside recorded periods are not deducted. Attribute each deduction to the affected task/area period; no wage calculation follows.
BR-021	Specific authority is required for corrections in every affected context. Retain current values without permanent field-change history. Conflicting accepted values require authorised comparison and renewed validation, not silent overwrite. Local proposals remain provisional and rejected proposals do not contribute to accepted central totals.
BR-022	Worker documents are contract, insurance or other. Upload requires same-business worker associations, category, title and PDF/JPG/PNG; observations are optional, timestamps/version identity automatic. The 10 MB per-file threshold remains proposed. Generation is conditional on UC-20 and is not implied by upload.
BR-023	A shared document has one source/version history and several same-business worker associations. Application signatures remain per worker/version and each worker-specific view shows that worker's captured signature. Source content may contain other information; individual-signature display does not redact it.
BR-024	Profile consultation does not grant all document access. Document consultation, modification and signing initiation are separate permissions. Relevant insurance consultation is mandatory within a manager's worker responsibility; contracts/other documents require additional authority. Financial access is separately controlled.
BR-025	Insurance validity reflects recorded expiry, not upload order: missing date is unconfirmed, passed expiry is expired and an unpassed expiry has valid recorded expiry status. It is not independent coverage verification. Expired retained files remain consultable and withdrawal is a separate restriction.
BR-026	A new accepted version becomes current without inherited signatures or external declarations. Contracts retain all versions; insurance/other documents retain current and immediate previous, deleting older signed or withdrawn versions after successful replacement. Ordinary deletion requires an unsigned dependency-free target and confirmation; signed errors are withdrawn while retained. A withdrawn current stays current and withdrawn until replaced, without fallback. Cleanup covers related source, derivative and signature data; failure/cancellation/retry does not trigger extra deletion.
BR-027	An external-signature declaration describes the uploaded version, not a verified application signing event or signatures for all associated workers.

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ID	Business rule
BR-028	Signing is in person on an authorised initiator's device. The initiator identifies the worker; the worker reviews document and drawing and explicitly confirms a non-empty signature. Redraw/cancel is allowed only before confirmation. Preserve worker, version, initiator and timestamp; do not replace or repeat a confirmed drawing for that pair. No worker account or owner signature is required.
BR-029	Prepare current/previous retained insurance for workers planned in the current/next week, with additional existing workers selected while connected. Other authorised versions may be explicitly prepared for offline signing; this does not automatically download all contracts. Preparation coverage does not define offline-session duration.
BR-030	A pending signature remains tied to the reviewed version. A newer version never inherits it. If the original remains eligible it may be accepted with a newer-version notice; withdrawal/deletion requires resolution and must not recreate a deliberately deleted version. Shared deletion reaches prepared copies on synchronisation, not instantaneously while disconnected.
BR-031	The proposed baseline is one transport per delivery in one campaign, with eligible source property/parcel entries, registered worker driver, vehicle, destination and departure timestamp. Collection origin/variety may be unknown and destination may be new. Confirm arrival not before departure, either later or retrospectively in one entry. Weights/results/amounts are not prerequisites and driver selection creates no attendance/hours. Independent trip management and external-driver support are not assumed.
BR-032	Each ticket belongs to one delivery; inverse multiplicity and source relationships remain unresolved. Weights retain meaning/unit and missing is not zero. Ticket reference is interpreted with source context. Adjustments retain known type/unit/application status; no unsupported deduction, inferred final weight, double deduction or assumed additive aggregation is allowed.
BR-033	Each analysis belongs to one delivery. Additional source analyses differ from field corrections; multiplicity/selection remain unresolved. Record reported yield percentage and known reference-weight meaning. No gross/net/payable reference is assumed. Pending results do not enter averages as zero and eligible weighted aggregation requires approved source rules.
BR-034	A ticket has at most one current cooperative-reported EUR amount, provisional or final. Final replaces provisional and corrections retain only current values. Do not add revisions, invent missing amounts, derive prices, reapply deductions or infer payment. Final-to-provisional reclassification is not assumed. Monetary source attachments are not worker documents.
BR-035	Mixed loads may contribute once to eligible campaign totals. Property/parcel allocation requires reported quantities/amounts or an approved rule; do not assign the full load to every source. Unallocated values remain identifiable and are not zero.
BR-036	Supported operational entry uses prepared offline data and current-value corrections. Monetary values/adjustments require online financial permission; mixed-content attachment handling remains unresolved. Local indicator creates count once, updates replace and removals suppress baseline contributions. Revalidate on reconnection and do not imply knowledge of other-device changes or accepted status for rejected proposals.

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ID	Business rule
BR-037	Review one campaign at a time. Participants are distinct workers with confirmed daily presence and actual campaign assignment, even with incomplete hours. Working days are distinct start dates with eligible actual periods. Hours use actual periods and same-worker breaks, by recorded task or unclassified; plans/capability do not establish actual totals. Distinct workers/dates are not additive across area breakdowns.
BR-038	Count distinct confirmed deliveries independently of tickets/analyses. Aggregate only eligible compatible weights, selected analyses and current ticket amounts. Identify missing/ambiguous data, partial coverage and unavailable calculations rather than displaying unsupported zeros. Theoretical oil is not oil actually received.
BR-039	Filters apply only to indicators with a defined source relationship and labelled date basis. Worker/task filters do not allocate deliveries or money. Underlying-record access remains separately authorised; partial scope is identified without revealing restricted information through detail or differences between totals.
BR-040	Campaign closure does not freeze current indicators: accepted late data and permitted corrections update derived totals. PDFs remain authorised timestamped snapshots with filters, completeness and synchronisation limitations. Financial content requires online financial access and is excluded from offline views/reports. Detailed spreadsheets and cross-campaign comparison remain outside baseline.

6.2 Record States and Transitions

6.2.1 Accounts, Invitations and Access

An account moves from *Pending verification* to *Active* after valid email verification. An expired link does not itself delete the pending registration. Active identity is separate from business membership.

Invitations move from *Pending* to *Accepted*, *Cancelled* or *Expired*. Only acceptance by the intended verified account activates membership. A manager membership moves from *Active* to *Revoked*; restoration requires a new explicit owner authorisation through invitation, not duplicate operational records.

A session is authenticated, expired or terminated. Offline access additionally requires prepared data and unexpired offline authorisation, renewed online. Account suspension/deletion and ownership transfer are not defined by these states.

6.2.2 Properties, Parcels and Workers

Properties/parcels move between *Active* and *Archived*; workers between *Active* and *Inactive*. Deletion is terminal and subject to dependency checks, not a reversible availability state. A parcel's availability also depends on its parent; parent reactivation does not undo independent parcel archival. Worker inactivation preserves future plans with warnings.

6.2.3 Campaign Lifecycle and Archival

Creation produces an *Open* campaign; closure produces *Closed*; reopening returns to *Open* after date and overlap validation. "Reopened" is not a separate permanent state. Planned/actual dates

distinguish expectation from activity without adding a mandatory in-progress state. Archival is independent: open or closed campaigns may be archived, without releasing coverage or itself changing operational permission. Deletion requires no operational records.

6.2.4 Attendance and Completeness

Attendance is *Awaiting confirmation*, *Present* or *Absent*, once per worker and working date in the business. Initial confirmation establishes presence/absence; later changes follow correction rules. Actual campaign-area assignment is separate. A late arrival uses actual times, not another attendance state. Work completeness is derived for the record/calculation; a confirmed period may support a labelled partial total. No mandatory daily-close action is introduced.

6.2.5 Synchronisation

Operations are *Pending*, *Synchronising*, *Accepted*, *Failed* or *Conflicted*. Distinguish server rejection from temporary transmission failure and uncertain outcome; check the existing operation before retry. Local confirmation is not server acceptance. An offline delivery may be delivered physically while its confirmation is pending.

6.2.6 Documents and Signatures

Version currency (current, previous or older retained contract version), withdrawal (available/withdrawn), recorded insurance expiry (valid/expired/unconfirmed) and signature information are independent. Missing expiry is unconfirmed; an unpassed recorded expiry is not independent verification of coverage. Expired and withdrawn retained files remain historically consultable, but withdrawal prevents operational use. A withdrawn current version remains current until replacement, without fallback.

An external-signature declaration is not an application event and does not establish individual signers. Application signing is tracked per worker/version: no confirmed signature, locally confirmed/pending, accepted, or resolution required. Transport failure is a synchronisation condition. Confirmed drawings are not replaced; document retention may nevertheless delete an older signed insurance/other version.

6.2.7 Deliveries and Source Information

Departure establishes *In transport*; arrival confirmation establishes *Delivered*. Retrospective registration may create Delivered directly with both timestamps. Ordinary timestamp correction does not reverse delivery; exceptional reversal/cancellation remains under Q-DA-06.

Weighing/analysis completeness is independent: *Not recorded*, *Incomplete* or *Available*. A supplied attachment does not itself establish complete or calculation-eligible structured data; field thresholds remain under Q-DA-08.

Reported amounts progress from *Not reported* to *Provisional* or directly to *Final*; final replaces provisional. Missing differs from reported zero. Final-to-provisional reclassification remains under Q-DA-09, not an assumed normal transition.

6.2.8 Indicators and Report Snapshots

Indicators are derived, not editable records. A result may be available, partial, unavailable, provisional due to pending local changes and/or based on last synchronisation. These conditions may coexist. Financial provisional classification is distinct from local pending status. Zero is valid only when supported by the available scope and evidence. A PDF is a timestamped snapshot; source corrections require a new export. Offline reports exclude monetary information and identify freshness and pending work.

6.3 Corrections and Change History

6.3.1 Identity, Dependencies and Current Values

Descriptive edits preserve record identities and associations; parcel creation does not distribute past property-level records. Ordinary profile/work/source-entry corrections retain current values rather than permanent field-change history. Distinct source analyses, document versions and signatures remain domain records, subject to their own retention. Security-event retention is a separate NFR decision; secrets must not be placed in event logs.

Coverage correction follows BR-010: reopen if closed, resolve dependent records, then remove the area. No cascade deletion or undefined exceptional reassignment is introduced. Campaign archival does not substitute for reopening.

6.3.2 Completion and Correction

Completion supplies previously unrecorded information without replacing or removing confirmed values or changing confirmed associations. It includes missing hours/tasks and previously unrecorded periods within valid actual dates. In a closed campaign it still requires attendance, coverage and overlap consistency. An absent worker cannot acquire actual work through completion; attendance must first be corrected. Confirmed work changes/removals require reopening, with no additional age-based deadline. Work outside valid dates requires reopening/date review. Existing-delivery tickets, analyses and amounts follow the separate BR-012 exception.

6.3.3 Work Corrections

Require correction permission and valid origin/destination campaign contexts. Worker, date, task, periods, breaks and area may be corrected without rewriting the plan. Moving a period preserves unrelated work and daily-attendance consistency. Presence-to-absence requires explicit removal confirmation for all contradictory actual assignments, including incomplete ones, periods and breaks on that date; all affected permissions and campaign reopening rules apply.

6.3.4 Access Revocation and Offline Conflicts

Pending proposals keep their origin account/business. Normal acceptance requires current authorisation. Existing revoked-manager proposals may undergo owner review under BR-004 without restoring access; the design must distinguish them from new unauthorised work. owner approval cannot bypass other business rules. Logout protects retained pending information from other accounts.

Concurrent accepted values require authorised comparison against the pending proposal; retain server values or submit a reviewed correction and validate again. Temporary conflict data is not

permanent correction history. Pending totals remain provisional and rejected proposals do not enter accepted totals.

6.3.5 Documents and Signing

Metadata correction must not overwrite signed contents. Replacing contents creates a new version; signatures and external declarations do not carry forward. Shared updates affect the same document across worker associations; hiding other application signatures does not redact the source file.

Ordinary deletion permits only an unsigned dependency-free incorrect upload; signed errors are withdrawn while retained. Rolling insurance/other retention is an explicit exception: after accepted replacement delete older source, derivative and signature data. No cleanup follows failure, cancellation or retry. Synchronise deletion to local copies and align storage/backups with the policy.

Pending signatures stay on the reviewed version. If it remains eligible, accept with notice of a newer version; if withdrawn/deleted, resolve under Q-DOC-04, never transfer the signature or recreate deleted content. Uncertain outcomes resume/check the existing signing operation, not another drawing.

6.3.6 Delivery-Related Corrections and Indicators

Ordinary transport corrections validate chronology and associations within the same campaign/state; exceptional cancellation, reversal and cross-campaign moves remain under Q-DA-06. Ticket/analysis corrections replace current fields, not genuinely additional source records. Amount corrections replace the single current valuation and require connectivity and financial permission. Source attachments remain distinct from correction history and worker documents.

Indicators cannot be corrected directly. Recalculate affected origin/destination contributions after source changes; local updates replace their baseline rather than adding it again. Closure does not freeze current results and existing PDFs remain unchanged snapshots. Changing a viewing filter does not change source data.

6.4 Calculation Rules

6.4.1 Areas, Dates and Eligibility

Declared hectares are optional; unknown is not zero. Parcel totals need not equal property area and no quantity allocation follows automatically from hectares. Future map estimates remain separate from the declared reference. Open campaigns reserve coverage under BR-008; dates on closed campaigns remain relevant. Working-date filters use period start dates; delivery counts use arrival dates. Ticket/analysis/financial date bases remain under Q-IND-01.

6.4.2 Counted Working Time

For a valid period with complete timestamps, raw hours are

$$D_i = \frac{t_{\mathrm{end},i} - t_{\mathrm{start},i}}{3600}.$$

The end is later than the start; an earlier clock time does not establish its date. Validate full intervals across adjacent dates. Let W_i be an eligible actual period and $w(i)$ its worker. Let B_w

be the union of applicable non-counted break intervals for that same worker. Then

$$h_i = \text{duration}(W_i) - \text{duration}(W_i \cap B_{w(i)}), \qquad H = \sum_i h_i.$$

Durations in this expression are in hours. A break affects only intersections with that worker's periods; the union prevents duplicate deduction. Counted breaks and already excluded gaps do not reduce hours. Task/area attribution uses the intersecting share of each period. Valid unknown-task hours are unclassified, not omitted. Planned or missing periods do not establish actual hours or zero. No wages are calculated. Rounding, time zones and break-entry validation remain under Q-WF-01–Q-WF-03.

Acceptance examples.

Input	Result
08:00–18:00 with no non-counted break.	10 hours.
08:00–18:00 with a counted 12:00–13:00 break.	10 hours.
08:00–18:00 with a non-counted 12:00–13:00 break.	9 hours.
08:00–12:00 and 13:00–18:00.	9 hours, without another deduction for the excluded gap.
08:00–11:00 machinery and 11:00–13:00 general work.	3 machinery hours and 2 general-work hours.
22:00–02:00 the next day.	4 hours, associated with the start date.
A and B work 08:00–12:00; only A takes a non-counted half-hour break.	A: 3.5 hours; B: 4 hours; total: 7.5 hours.
A valid two-hour period with an unknown task.	2 unclassified hours.

6.4.3 Campaign Work and Delivery Indicators

Let P_C be distinct workers with confirmed presence and actual assignment to campaign C; let E_C be eligible actual periods, with start date d_i. Let D_C be distinct confirmed deliveries in scope. Then

$$N_{\text{workers}} = |P_C|, \qquad N_{\text{days}} = |\{d_i : i \in E_C\}|, \qquad N_{\text{deliveries}} = |D_C|.$$

Participation can be known without hours; a no-period assignment does not add a measured working day. Driver selection alone is not participation.

$$H_C = \sum_{i \in E_C} h_i, \qquad H_{C,w} = \sum_{\substack{i \in E_C \\ w(i)=w}} h_i, \qquad H_{C,f} = \sum_{\substack{i \in E_C \\ \text{task}(i)=f}} h_i.$$

Task totals use recorded tasks and identify unclassified hours. Distinct workers and days cannot be summed across area breakdowns. Incomplete intervals support no invented duration; partial known totals disclose missing coverage.

6.4.4 Weights and Yield

For eligible non-duplicated records E_w with compatible units and the same approved weight meaning, $W_{\text{total}} = \sum_{i \in E_w} w_i$. Multiple tickets do not establish additive quantities. Do not infer final weight or apply deductions until types, bases, already-applied status, order and rounding are confirmed under Q-DA-01 and Q-DA-02.

For eligible analyses E_a, a candidate weighted percentage is

$$\bar{r} = \frac{\sum_{i \in E_a} w_i r_i}{\sum_{i \in E_a} w_i},$$

with reported percentage r_i and approved reference weight w_i. This model remains a proposal under Q-DA-03 and Q-DA-04; it is unavailable with no eligible positive denominator. Missing results are not zero and partial coverage is disclosed. Theoretical oil remains distinguishable from oil actually received and its inclusion is not assumed.

6.4.5 Current Monetary Totals and Allocation

For eligible tickets with current final amounts E_f and provisional amounts E_p, respectively:

$$V_{\text{final}} = \sum_{t \in E_f} a_t, \qquad V_{\text{provisional}} = \sum_{t \in E_p} a_t.$$

Use one current EUR amount per eligible ticket and decimal monetary precision. Replaced values contribute nothing; missing amounts are disclosed, not zero. Do not sum multiple tickets without confirmed semantics or reapply deductions. Mixed-load totals contribute once to eligible campaign results; source-area allocation requires reported quantities/amounts or an approved rule. Otherwise show unallocated information, not duplicate totals or invented zeros.

6.4.6 Insurance, Retention and Reports

Recorded expiry compares the expiry date with the current date consistently online/offline: missing is unconfirmed, passed is expired and unpassed has valid recorded expiry status, without independent coverage verification. After accepted v3, contracts retain v1/v2/v3; insurance/other retain v2/v3, deleting v1 even if signed. Failed/cancelled v3 leaves v1/v2 unchanged and shared associations do not create separate retained copies. The proposed 10 MB upload limit and exact byte threshold remain under Q-DOC-06.

Recalculate indicators from current eligible sources under FR-061, preserving accepted/local distinctions. PDF snapshots identify scope, filters/date basis, generation timestamp, incomplete/unallocated data, last synchronisation and included pending changes. Offline output excludes money. No defined denominator, missing local information or failed retrieval is silently displayed as zero.

7 Non-Functional Requirements

7.1 Performance

7.2 Usability and Accessibility

7.3 Security

Password policy, authentication request limits, session expiry and verification, invitation and password-reset lifetimes remain to be specified. Security-event retention must also be defined. No numerical values are assumed at this stage.

7.4 Privacy

7.5 Reliability and Recovery

7.6 Availability and Offline Continuity

The maximum offline authorisation duration and the handling of expiry with pending changes remain to be specified. Renewal requires online validation. Reconnection revalidates current business access and permissions before pending operations are accepted.

8 Solution Architecture

8.1 System Context

8.2 Components and Responsibilities

8.3 Main Interaction Flows

8.4 Offline Operation and Synchronisation

8.5 Deployment and Operation

8.6 Architecture Decisions

9 Data Architecture

9.1 Conceptual Data Model

9.2 Logical Data Model

9.3 Data Dictionary

9.4 Integrity Constraints

9.5 Data Lifecycle and History

9.6 Local and Central Data

9.7 Physical Data Model

10 Validation and Decision Status

10.1 Validation Approach

Validation will use the acceptance criteria in Section 5.1 and the use-case mapping in Section 5.2. Each check must record the requirement and relevant lettered acceptance criterion (e.g. FR-041(a)), expected result, observed result and evidence. Documented behaviour, implemented functionality and successful verification are distinct; no implementation result is asserted here.

Review will cover the agreed end-to-end workflow and relevant alternatives, including offline entry, synchronisation, permissions and corrections. Pending business rules must be resolved before dependent calculations or behaviours can be accepted.

10.2 Key Decisions and Assumptions

The following decisions define the current working specification. Formal team or client approval has not yet been recorded here. Changes require coordinated updates to the affected use cases, requirements and rules. This summary does not replace their details.

ID	Current decision	Main references
D-01	One owner workspace contains the owner's properties, campaigns, worker records and associated information.	UC-01; FR-001; BR-001.
D-02	An open campaign reserves its areas, including against campaigns with future planned starts. Different parcels may coexist; archiving does not release coverage.	UC-06; FR-019; BR-008.
D-03	Actual start is entered manually or proposed from the first confirmed actual campaign-area assignment. It requires explicit, connected confirmation; presence or planning alone is insufficient.	UC-06, UC-10; FR-016; BR-009.
D-04	Missing work may be completed within a closed campaign's valid actual dates without replacing confirmed information. Changing or removing confirmed work requires reopening.	UC-10, UC-11; FR-035–FR-036; BR-011.
D-05	Ticket information, analyses and reported amounts for existing deliveries or tickets may be completed or corrected after closure without reopening.	UC-16–UC-18; FR-054; BR-012.
D-06	Reopen a closed campaign before coverage correction. Resolve dependencies before removing an area; unresolved records block removal and no cascading deletion occurs.	UC-06; FR-015; BR-010.
D-07	A withdrawn current document remains current and withdrawn until replaced. An earlier version does not automatically become current.	UC-13; FR-041; BR-026.
D-08	If managers are included, the owner may review existing pending proposals after revocation, subject to normal validation. Originating and accepting accounts remain distinct.	UC-04; FR-008; BR-004.

10.3 Pending Decisions and Their Impact

The following matters remain unresolved. Team review selects the proposal and assigns follow-up; client confirmation is needed where the answer depends on actual business practice. Record the responsible person, target date and validation evidence when agreed.

Topic and IDs	Decision and impact	Validation needed
Manager role Q-AA-01, Q-AA-02	Confirm inclusion, responsibility scope and permissions. Affects delegated workflows, access checks and delivery scope.	Team; client confirms operational responsibilities.
Account lifecycle Q-AA-07	Decide whether email changes, account/business deletion and ownership transfer are included. Affects lifecycle and data handling.	Team scope decision; client if operationally required.
Tickets and weights Q-DA-01, Q-DA-02, Q-DA-08	Confirm ticket multiplicity, weight meanings, deductions, calculation order, units, rounding and completeness using source examples. Blocks final source relationships and derived weights.	Client examples and explanation; team formalisation.
Analyses Q-DA-03, Q-DA-04	Confirm multiple-result selection and reference weights, yield formulas, precision and valid ranges. Blocks definitive yield aggregation.	Client; team validates calculations.
Delivery policies Q-DA-05–Q-DA-07, Q-DA-09, Q-DA-10	Confirm mixed-load allocation, exceptional correction/cancellation, registered-only drivers, one transport per delivery and final-to-provisional reclassification. Affects provenance, dependencies and totals.	Team proposal and client confirmation.
Documents and signing Q-DOC-01–Q-DOC-04	Confirm generation templates/data/output and offline scope, intended signing use, version retention and resolution of pending signatures after withdrawal/deletion. Affects UC-20 scope and document lifecycle.	Team; client confirms document needs and policies.
Offline access and security Q-AA-04, Q-AA-05, Q-DOC-07, Q-OFF-01, Q-OFF-02	Agree required offline operations, authorisation/session limits, prepared-file protection, revocation/deletion and online-only monetary boundaries, including mixed-content attachments. Affects NFRs, synchronisation and local data design.	Team; client confirms field needs.
Time and recovery Q-WF-01–Q-WF-04	Define display precision, time zones, break validation, existing-target-plan handling and recovery of conflicts/rejections. Preserve the agreed formulas and manual resolution rules.	Team; client where conventions depend on business practice.
Files and attachments Q-DOC-06, Q-AT-02	Confirm the proposed 10 MB worker-document threshold and separate source-attachment limits, retention and interrupted-upload recovery. Affects storage and transfer design.	Team; client validates representative files.

Topic and IDs	Decision and impact	Validation needed
Indicators and reports Q-IND-01, Q-IN-02	Select source-date bases, offline preparation/detail and PDF behaviour. Affects filtering, reproducibility and locally available reporting.	Team proposal; client validates reporting meaning.
Delivery and presentation Q-SCOPE-01, Q-UI-01	Agree first-delivery priorities and whether the 6 November 2026 presentation includes wireframes or a clickable prototype. Affects estimates, responsibilities and demonstration scope.	Team, within course requirements.

Unresolved matters must remain visible assumptions or limitations, not silently become approved requirements. The detailed meeting questionnaire and action assignments are maintained separately. Validated outcomes will be reflected in this summary and in the affected specification. Detailed meeting records remain separate from the submitted report.